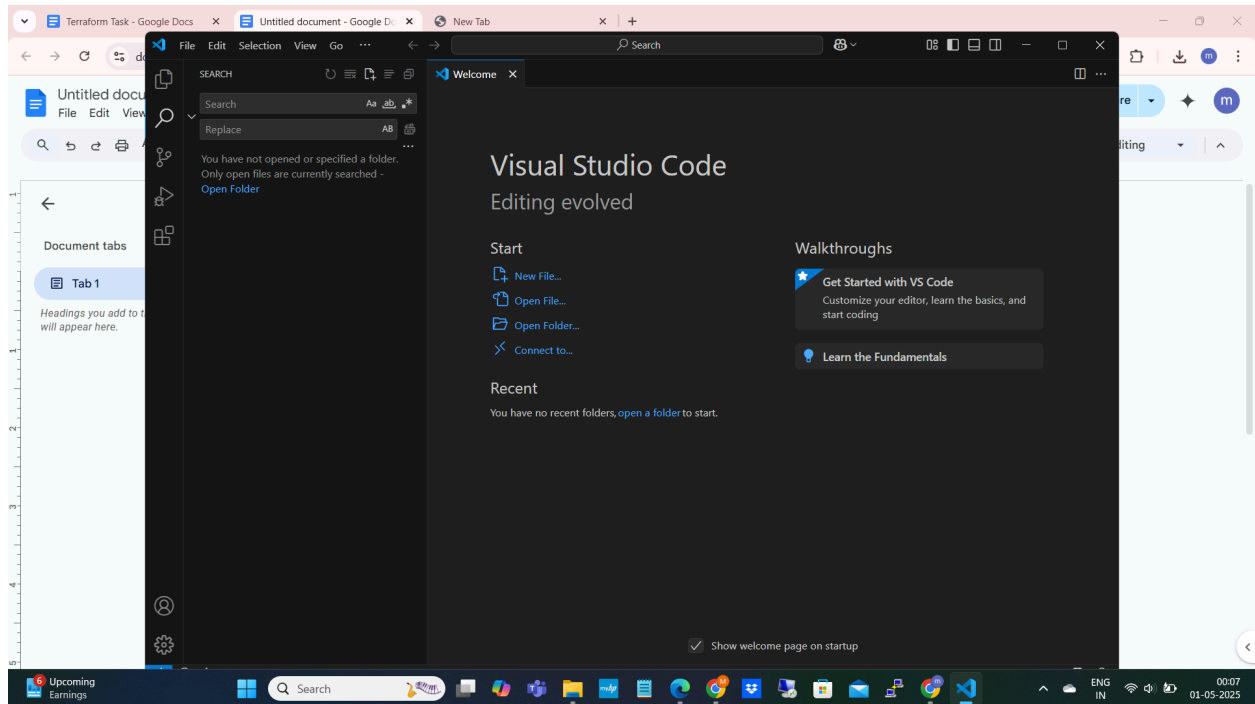


Terraform Task

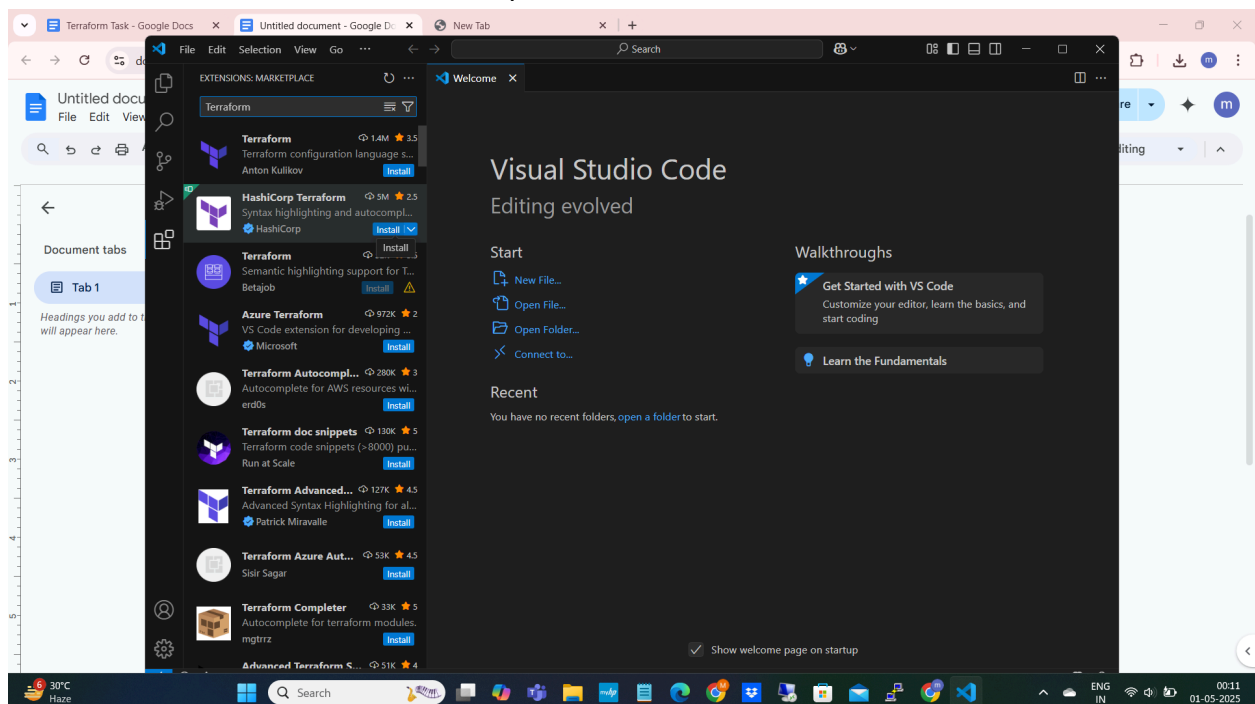
Installed Visual Studio

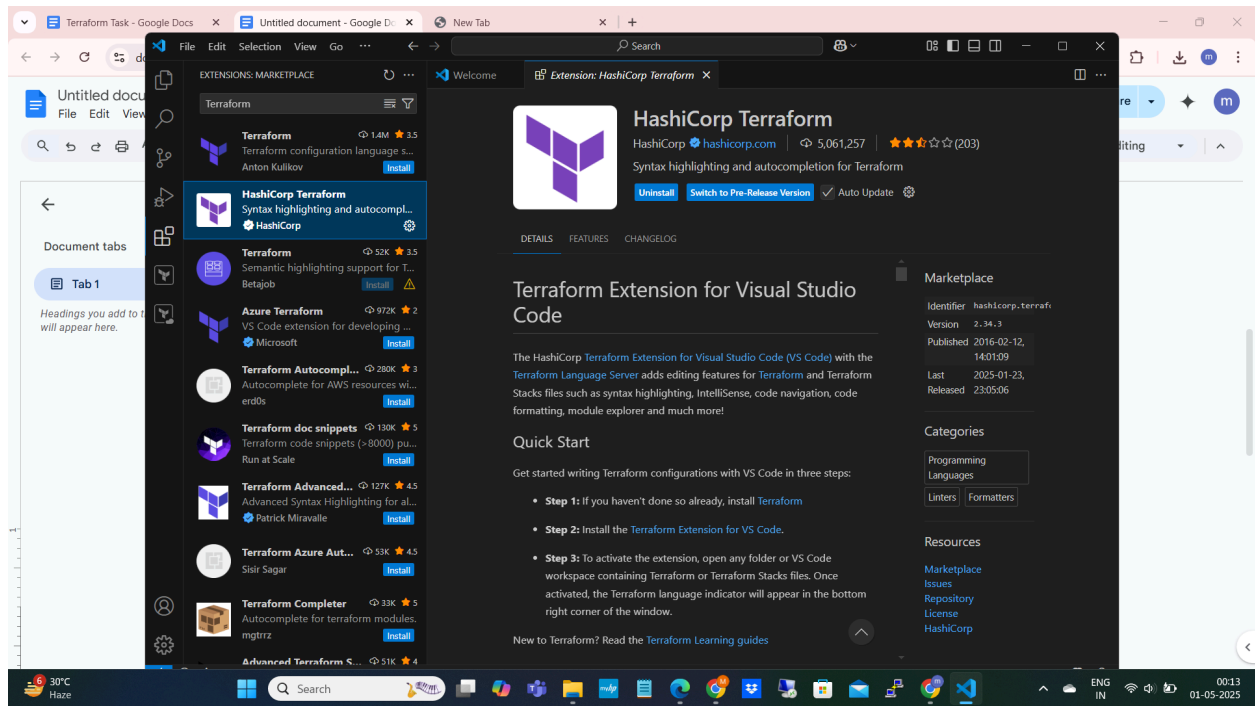


Install Terraform Extension:

Go to Extensions tab and Search for the Terraform

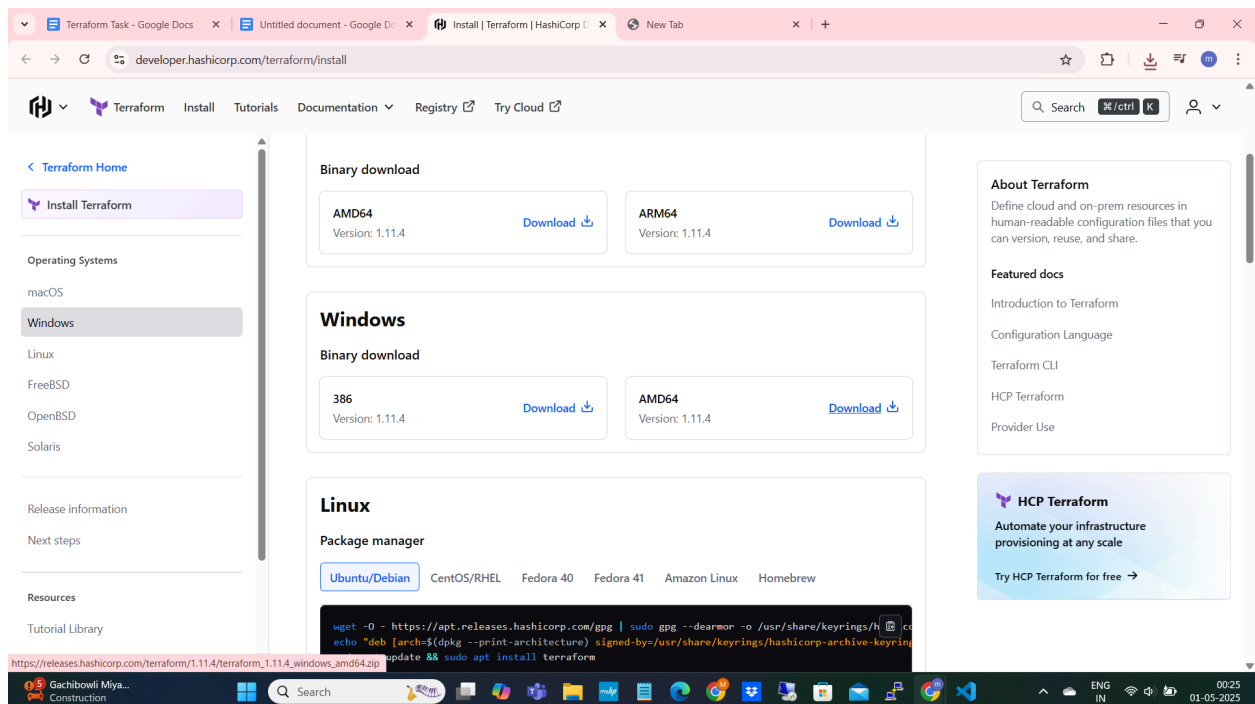
Install the extension from the HashiCorp



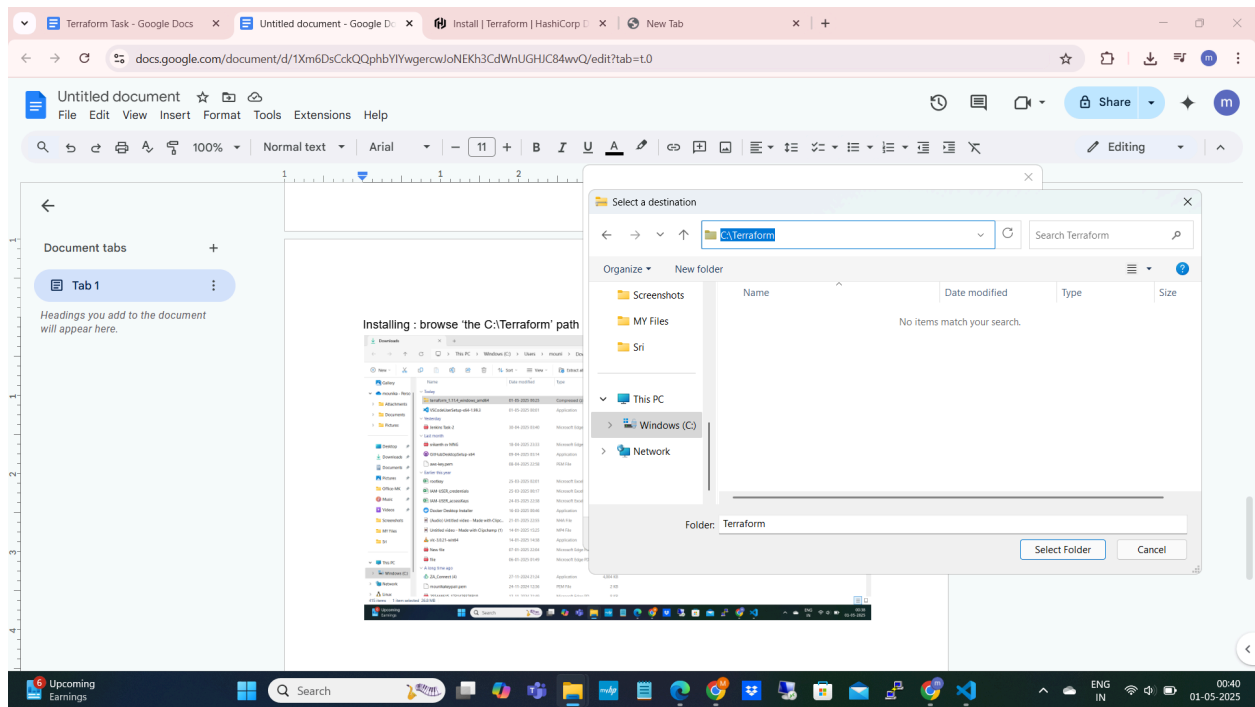
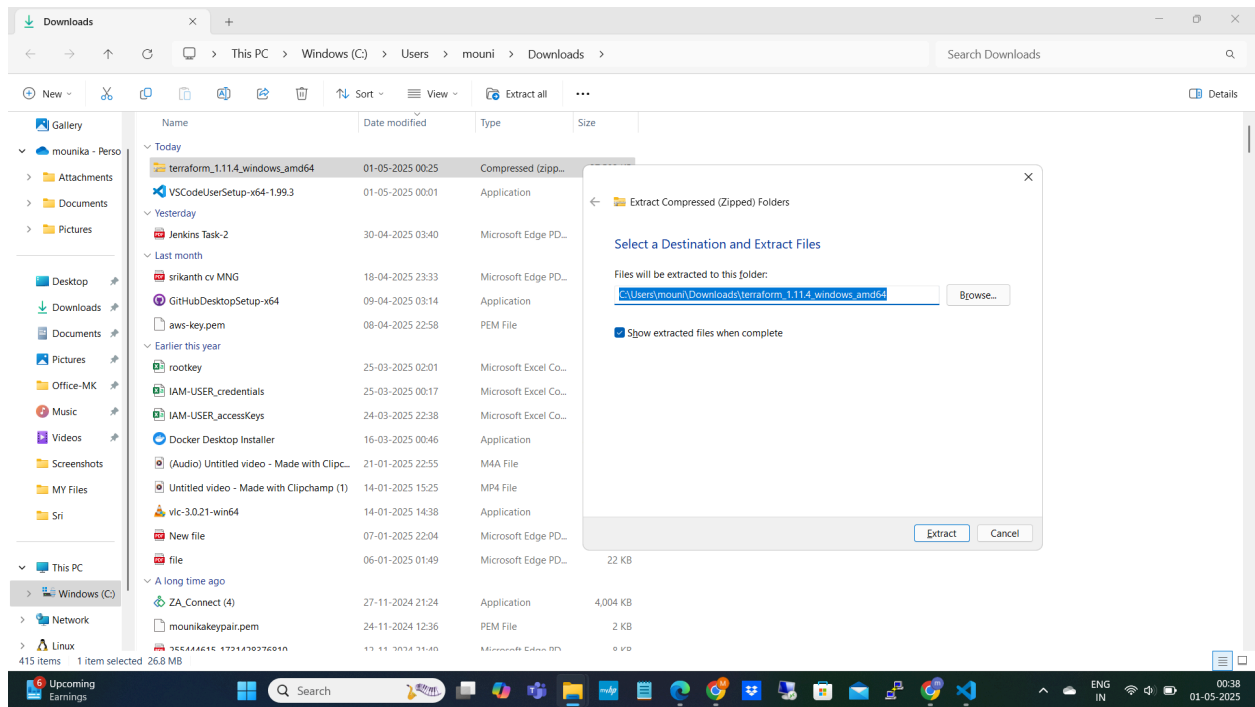


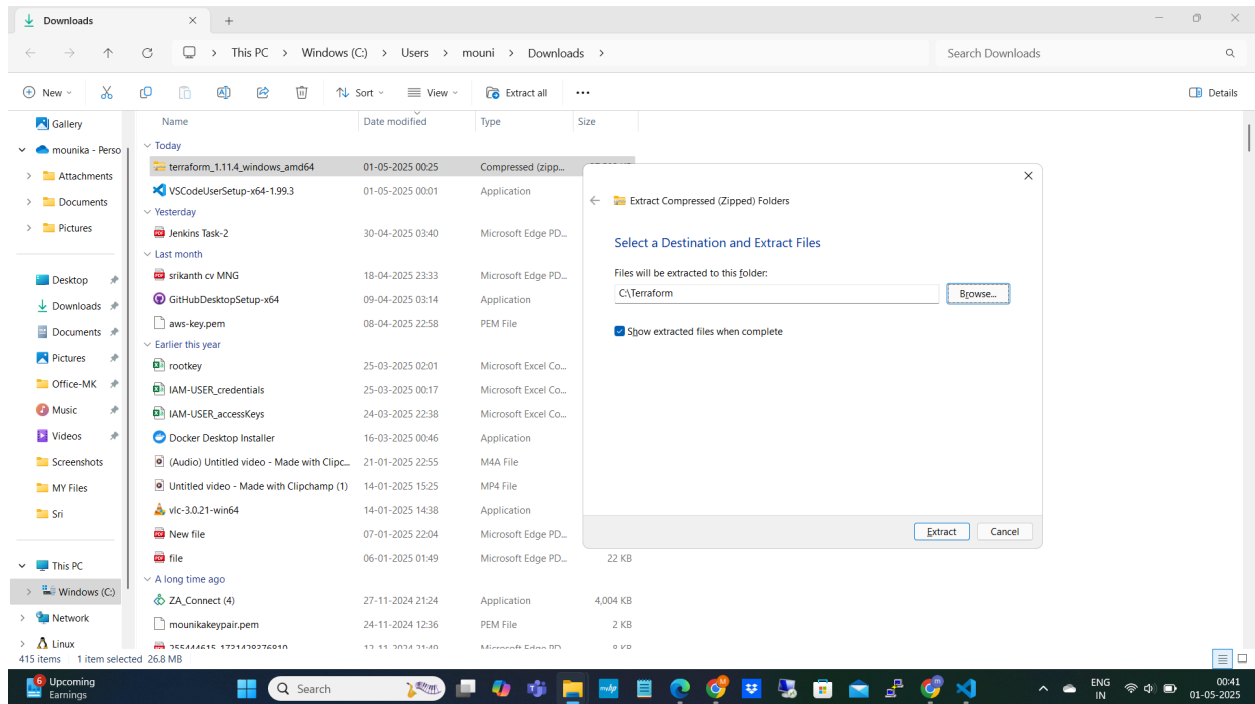
Installing Terraform in My Local machine:

Down loading the Terraform.exe file

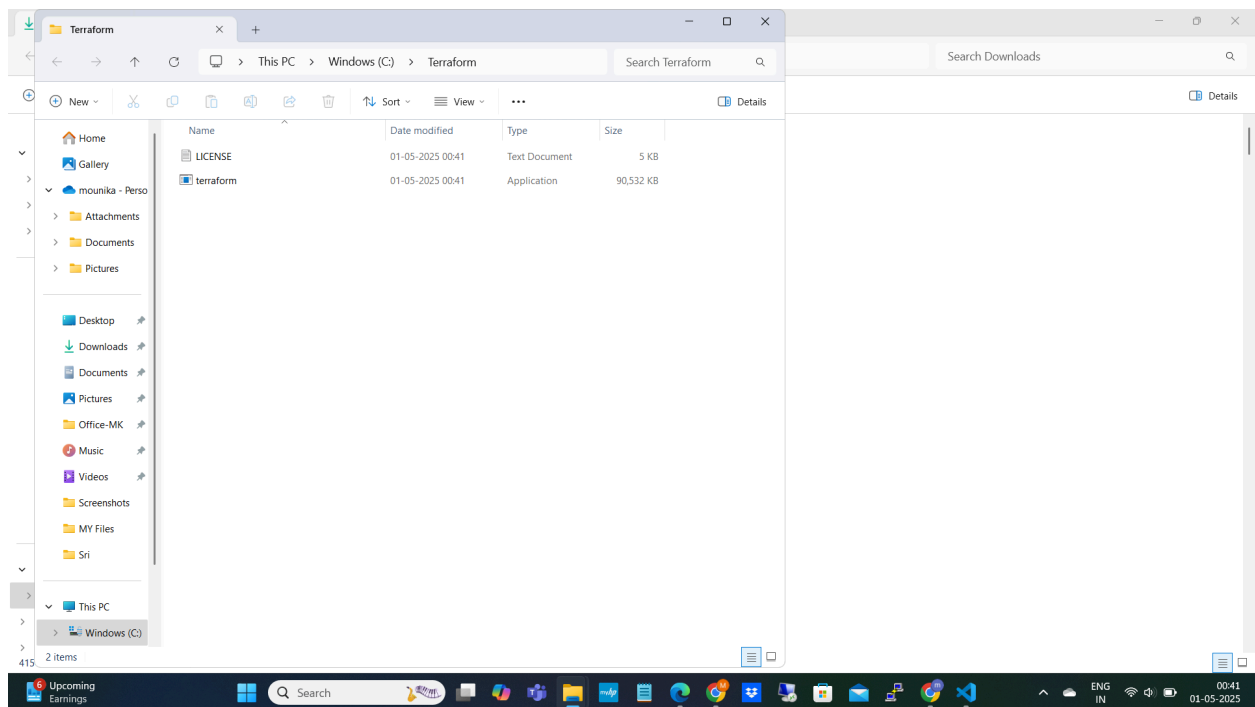


Installing: browse 'the C:\Terraform' path



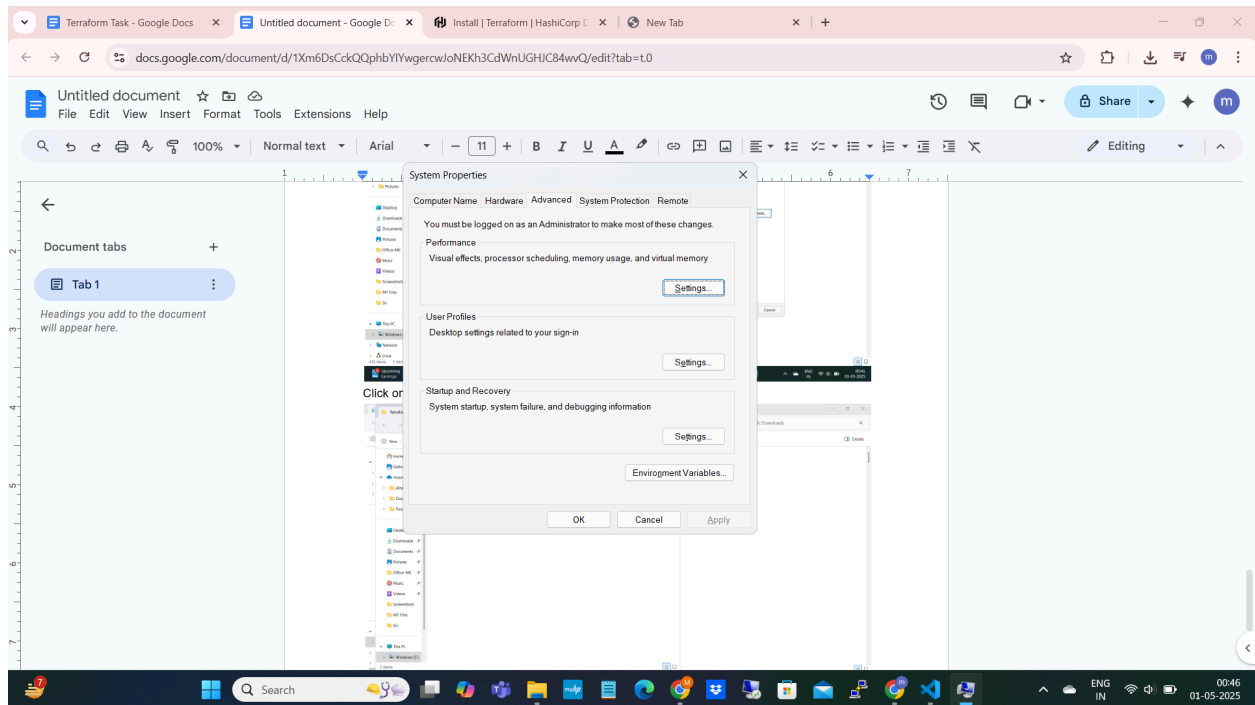


Click on Extract

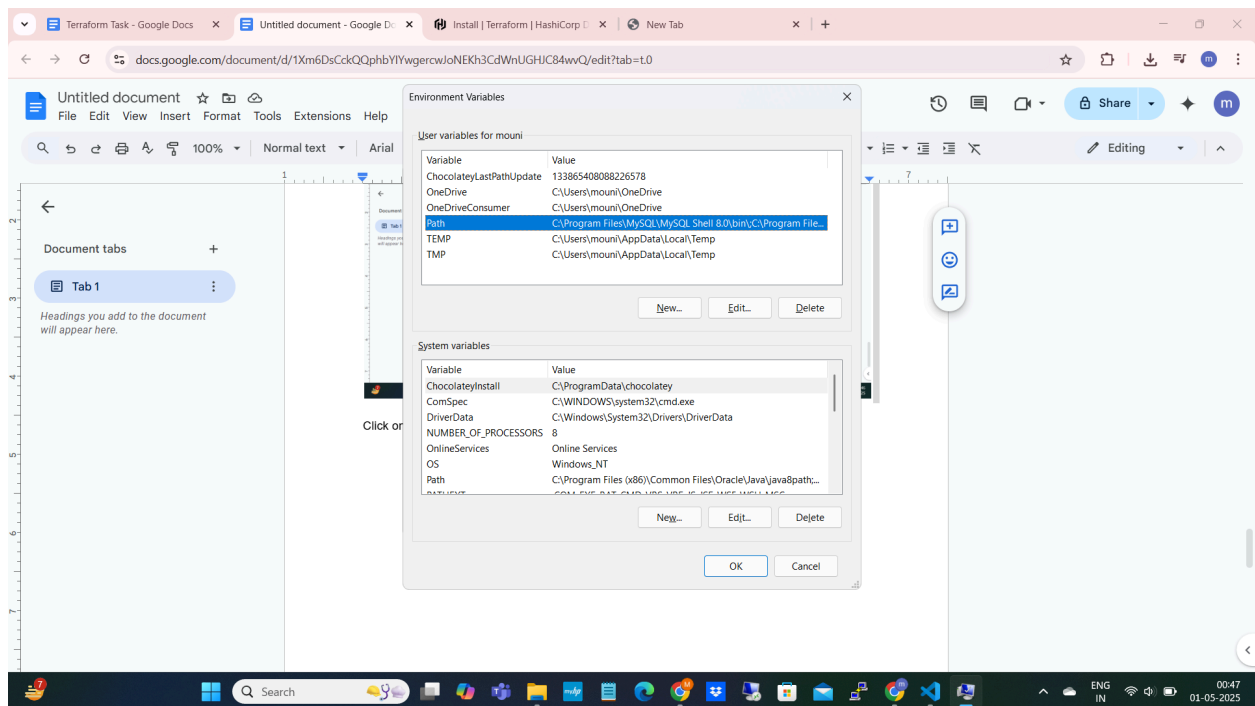


Now Set Environment Variable

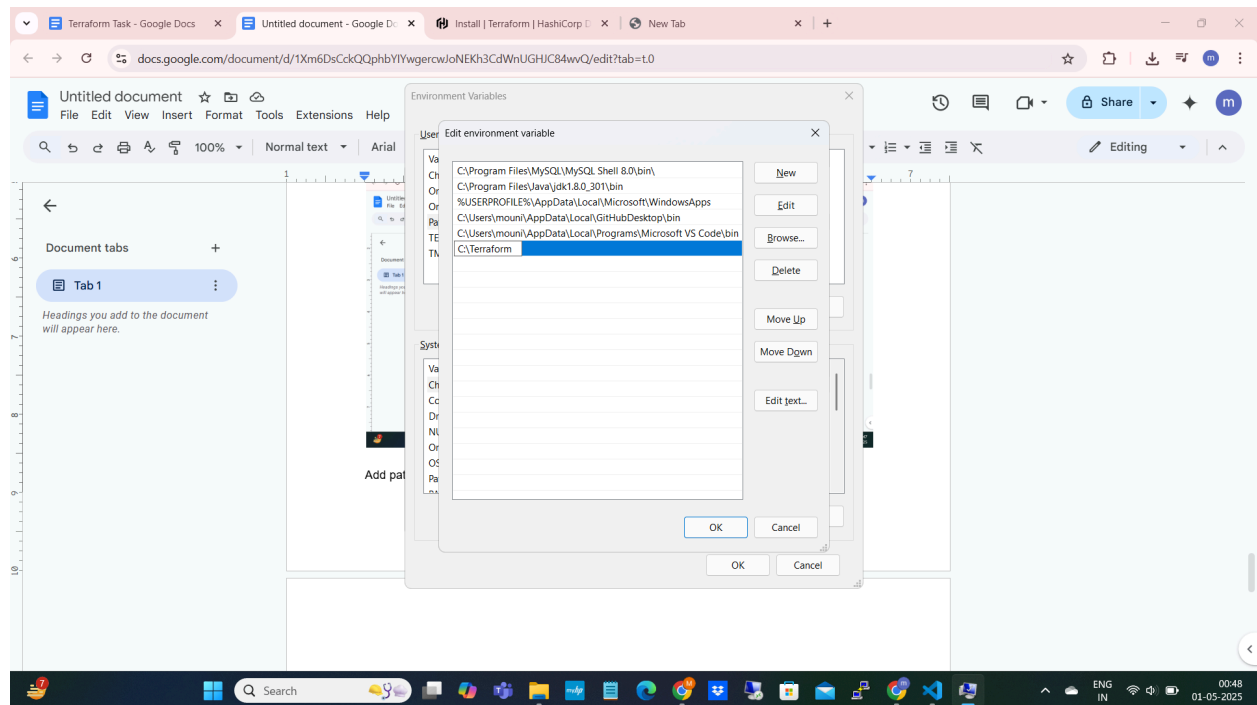
Advanced system settings:



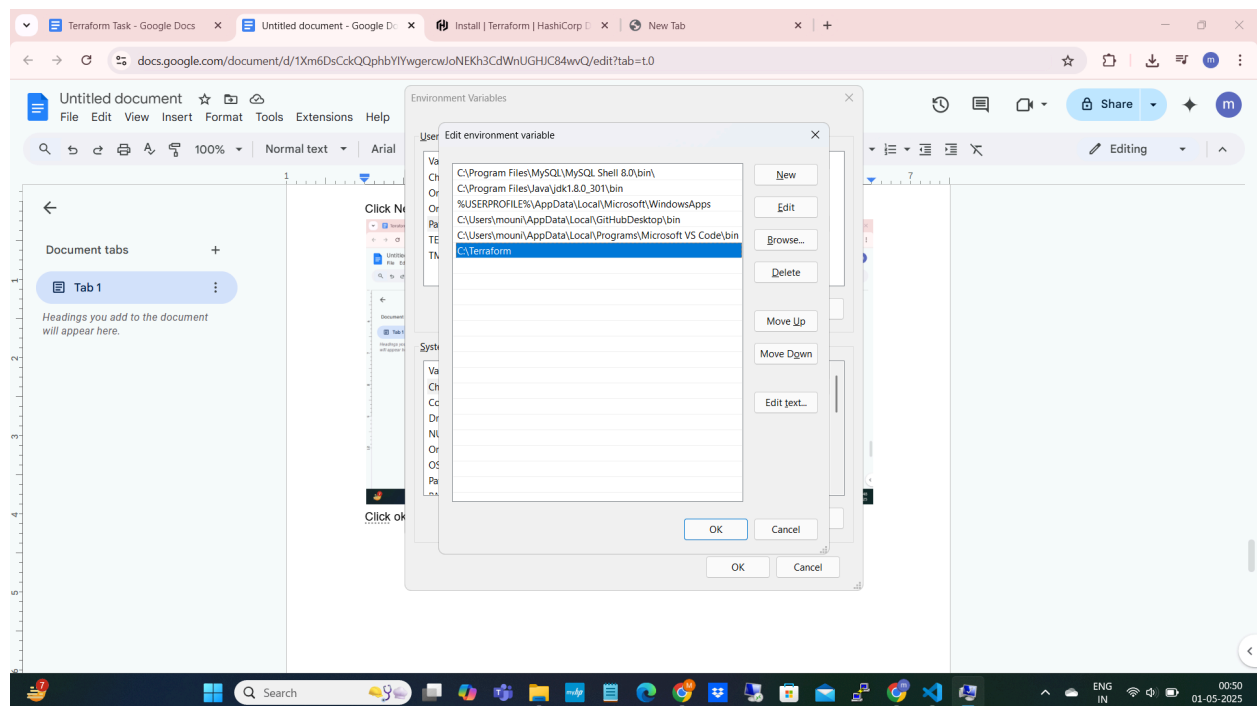
Click on Environment variables and Select the the path variable



Click New and Add the path **C:\Terraform**



Click ok



Then click ok

Verify installation

\$ terraform -v

```
C:\> Select Command Prompt

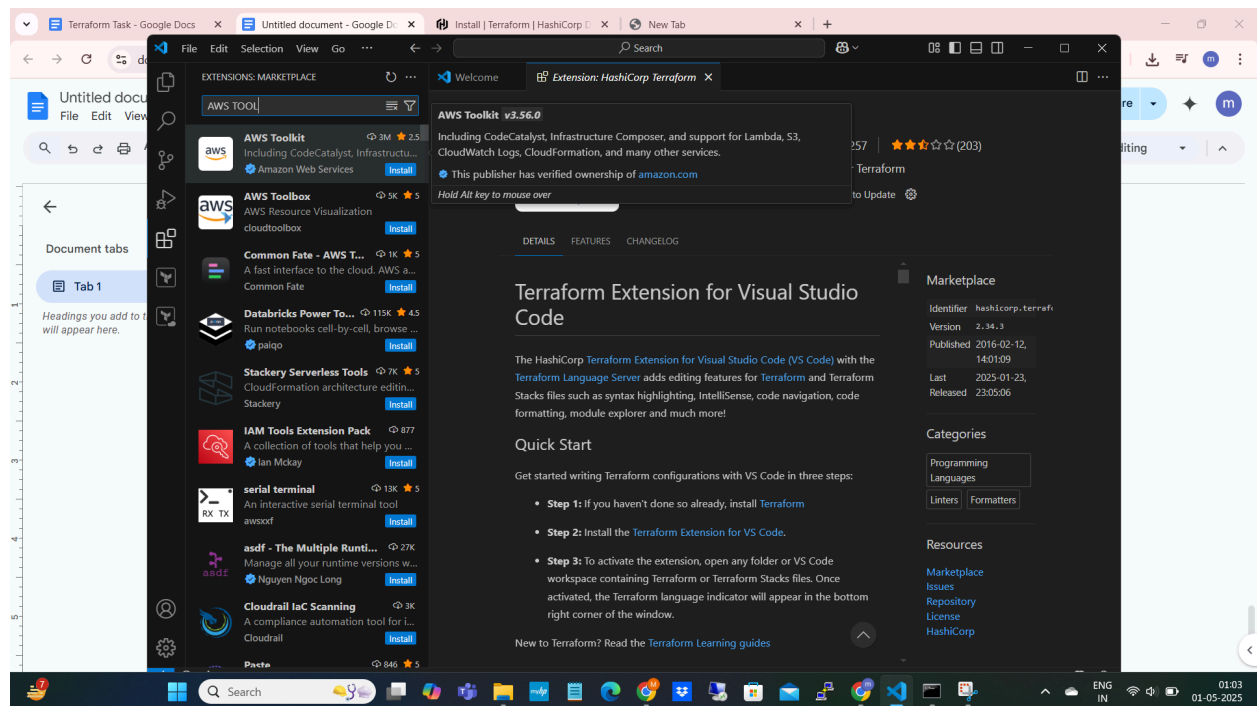
Microsoft Windows [Version 10.0.26100.3915]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mouni>terraform -v
Terraform v1.11.4
on windows_amd64

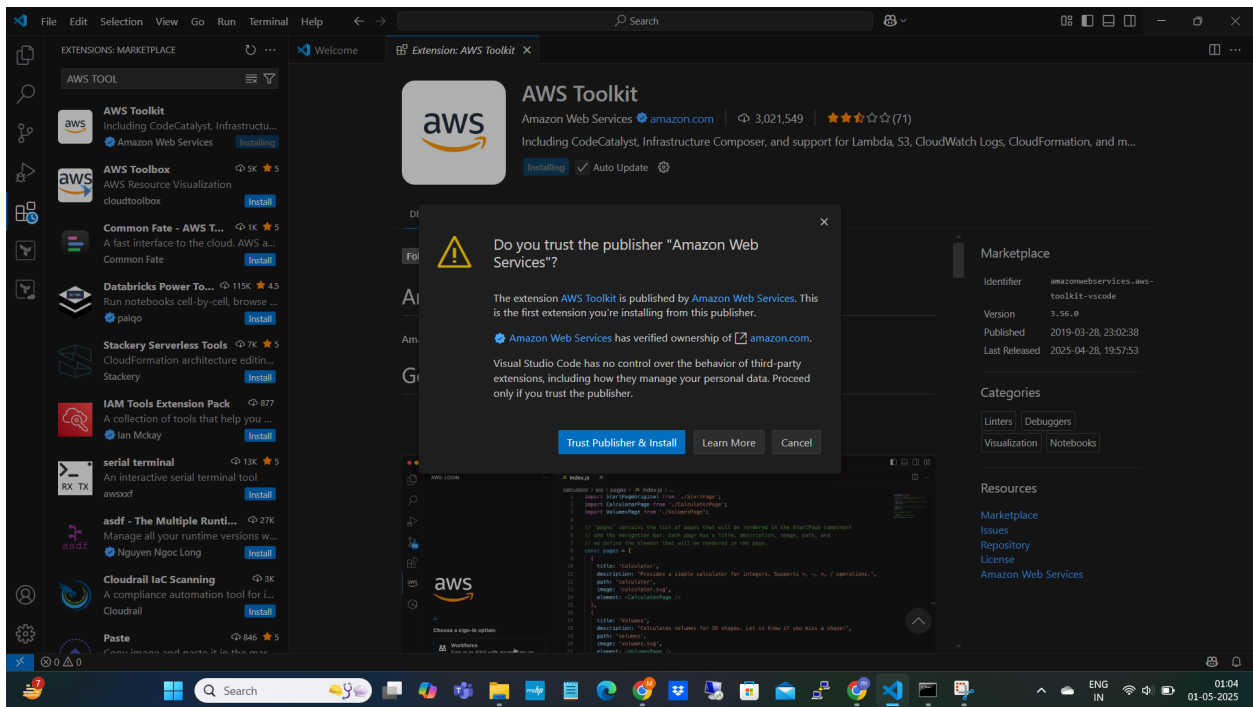
C:\Users\mouni>
```

Configure AWS Toolkit for VS Code:

Search for AWS toolkit in visual studio and select it



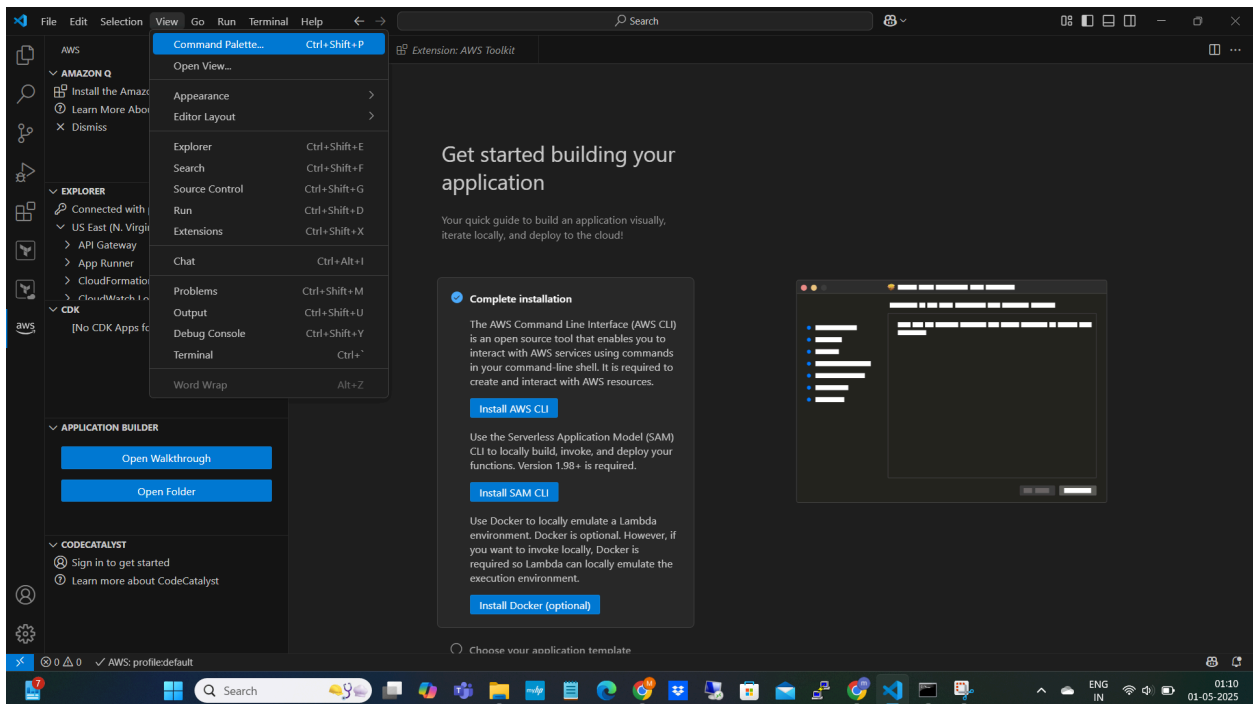
Click on Trust Publisher & install



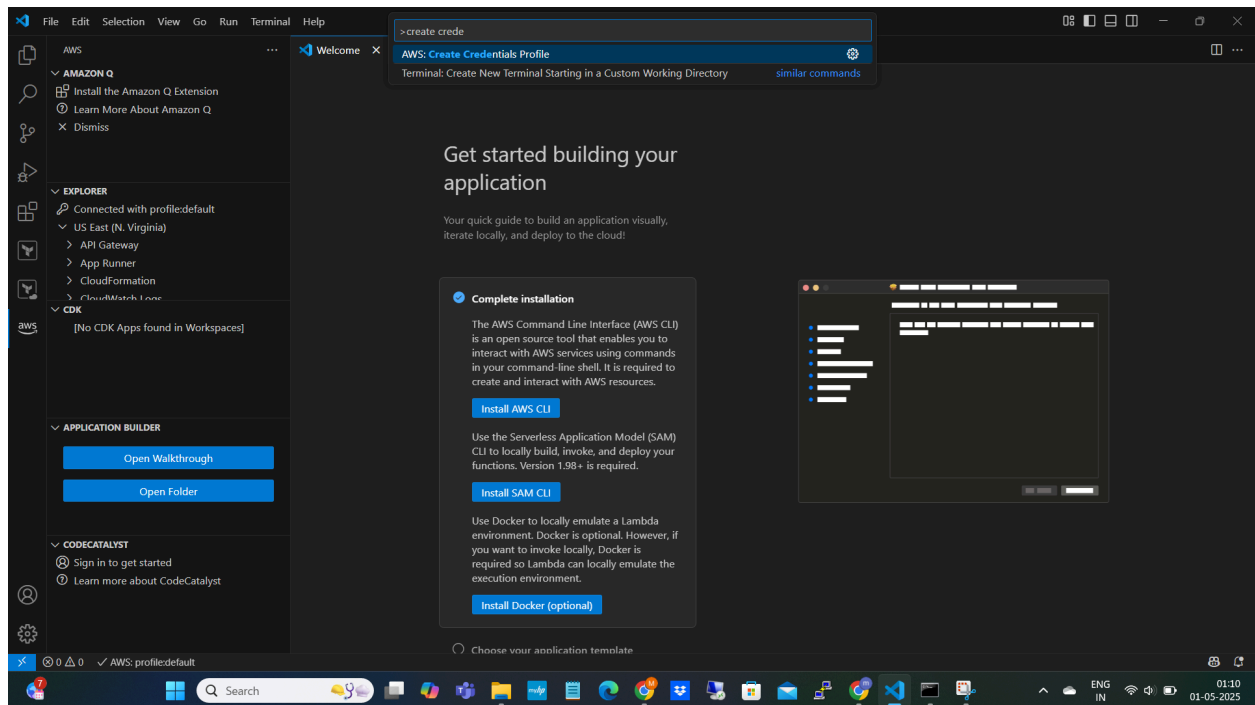
Installation done

Now Configure the AWS account credentials

Click on AWS icon from left panel →View→Command palette→Create credentials profile



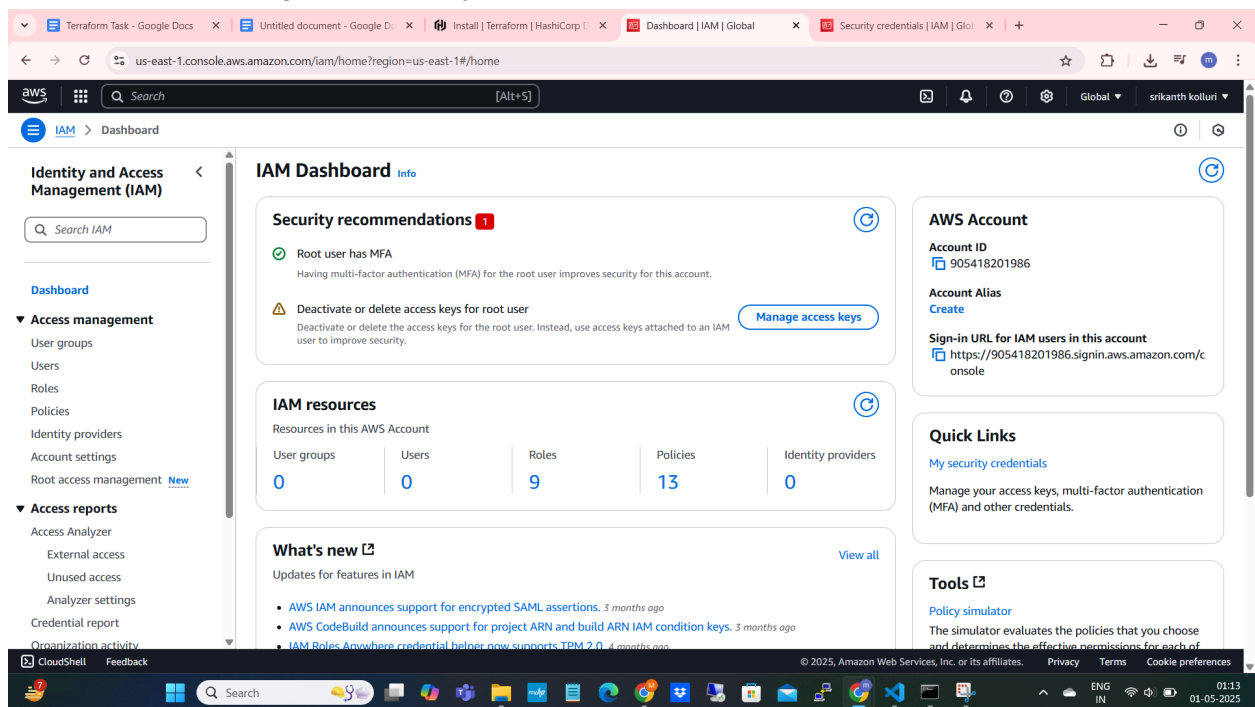
Create credentials Profile



We Need to create Access key and Secret Keys

Login to the AWS account and create

Click on IAM →Manage Access keys



us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/security_credentials

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management

Access reports

- Access Analyzer
- External access
- Unused access
- Analyzer settings
- Credential report
- Organization activity

Multi-factor authentication (MFA) (1)

Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. [Learn more](#)

Type	Identifier	Certifications	Created on
☐ Virtual	arn:aws:iam::905418201986:mfa/Authapp	Not Applicable	Sun Nov 24 2024

Access keys (1)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

Access key ID	Created on	Access key last used	Region last used	Service last used	Status
☐ AKIA5FTZBLOBFWX4POPW	36 days ago	9 minutes ago	us-east-1	sts	Active

CloudFront key pairs (0)

You use key pairs in Amazon CloudFront to create signed URLs. You can have a maximum of two CloudFront key pairs (active or inactive) at a time.

Creation time	CloudFront key ID	Status
---------------	-------------------	--------

No CloudFront key pairs

[Create CloudFront key pair](#)

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01:15 01-05-2025

Create Access key

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/security_credentials/access-key-wizard

IAM > Security credentials > Create access key

Step 1 Alternatives to root user access keys

Step 2 Retrieve access key

Alternatives to root user access keys [info](#)

Root user access keys are not recommended

We don't recommend that you create root user access keys. Because you can't specify the root user in a permissions policy, you can't limit its permissions, which is a best practice.

Instead, use alternatives such as an IAM role or a user in IAM Identity Center, which provide temporary rather than long-term credentials. [Learn More](#)

If your use case requires an access key, create an IAM user with an access key and apply least privilege permissions for that user. [Learn More](#)

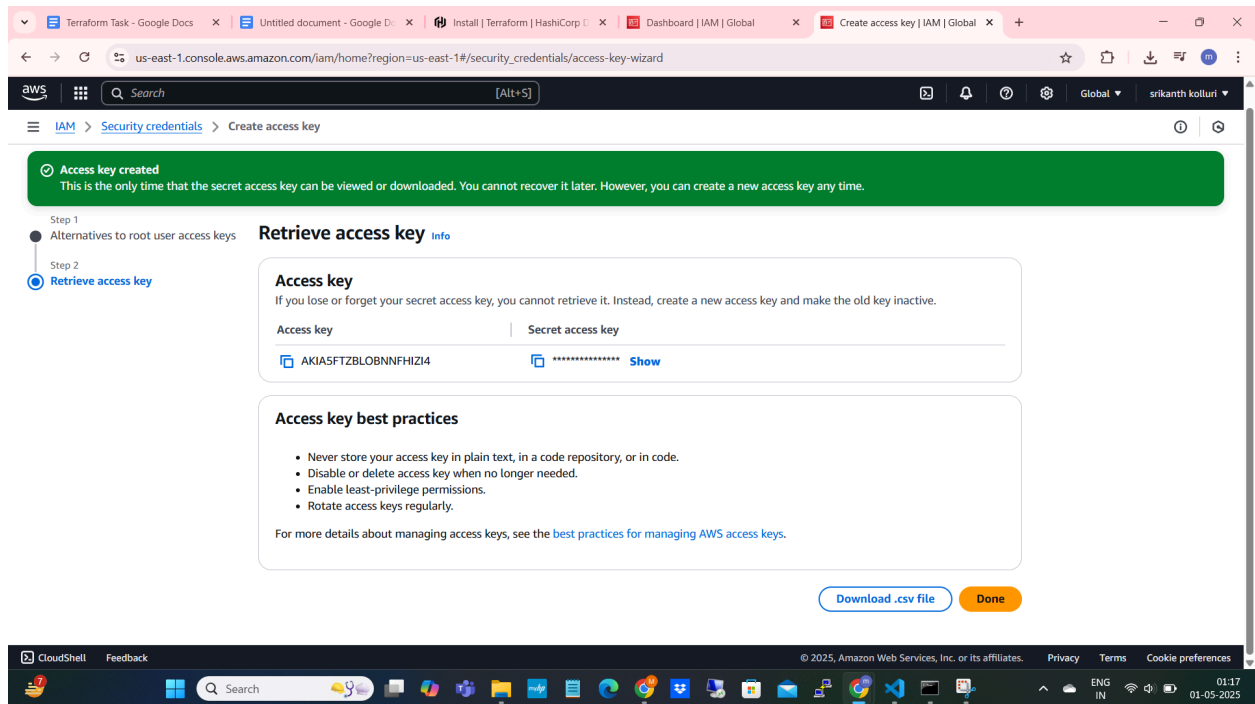
Continue to create access key?

☒ I understand creating a root access key is not a best practice, but I still want to create one.

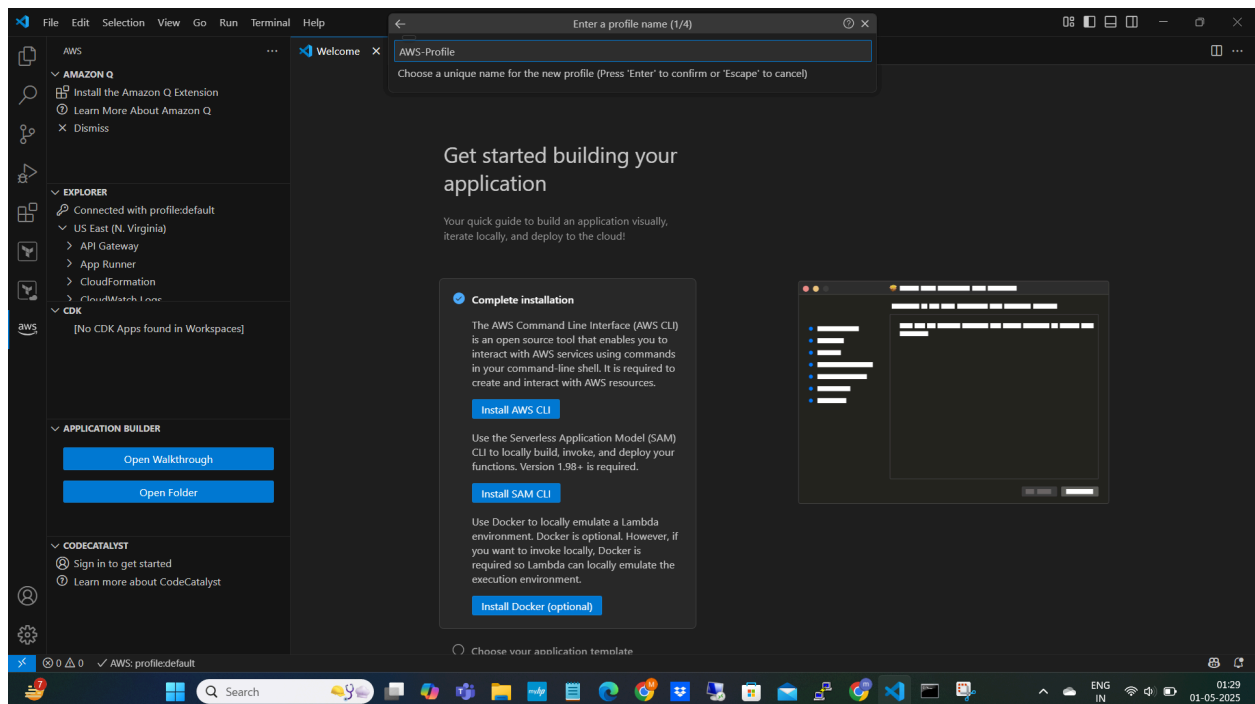
[Cancel](#) [Create access key](#)

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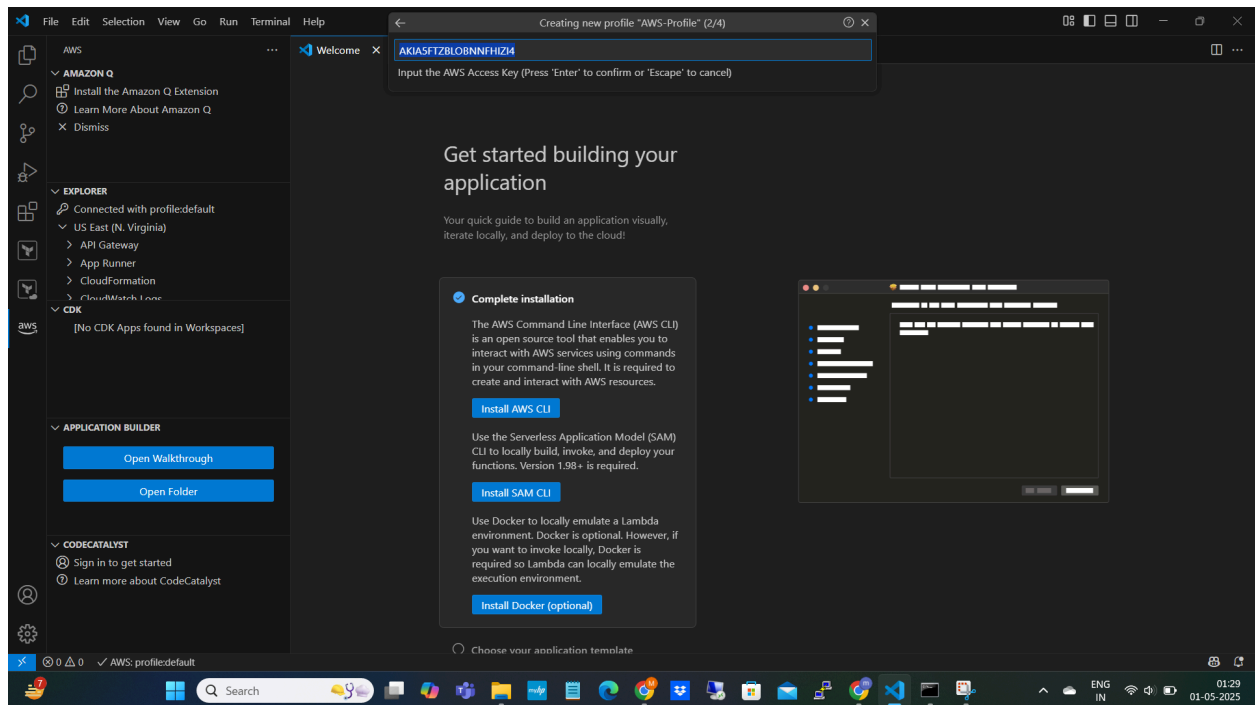
01:17 01-05-2025



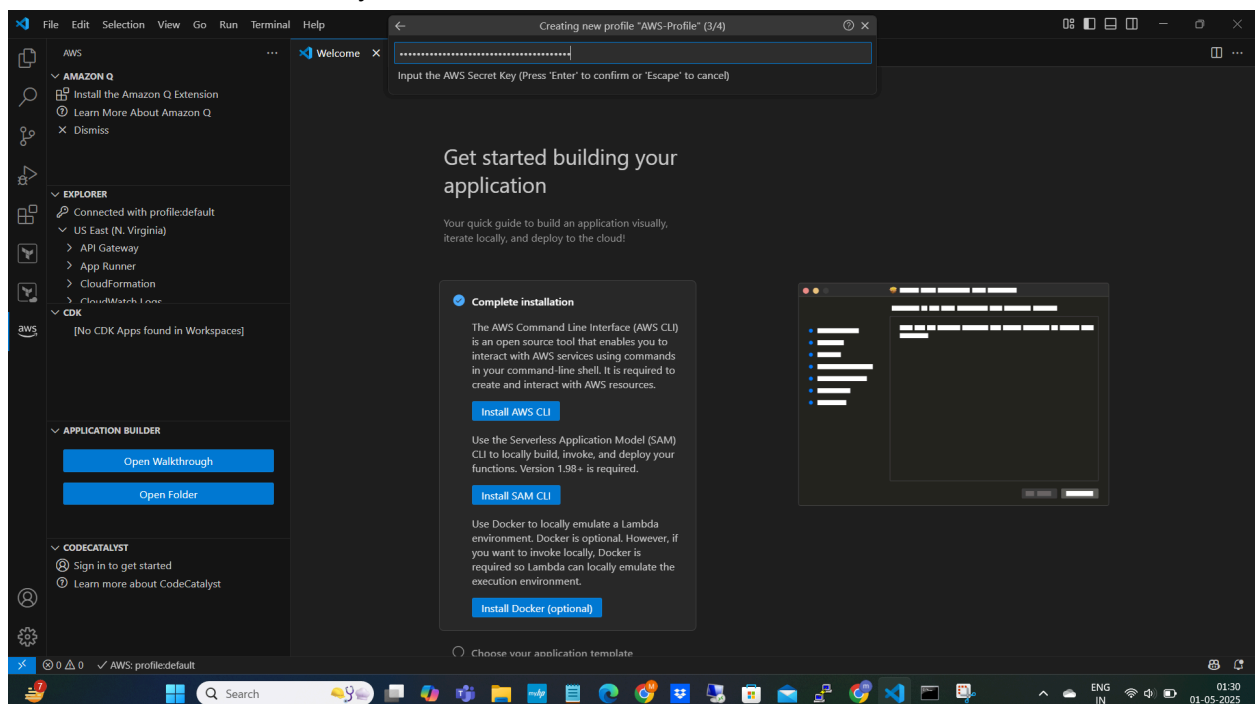
To connect AWS we must configure the access key and Secret key in vscode
Enter Unique name and click Enter

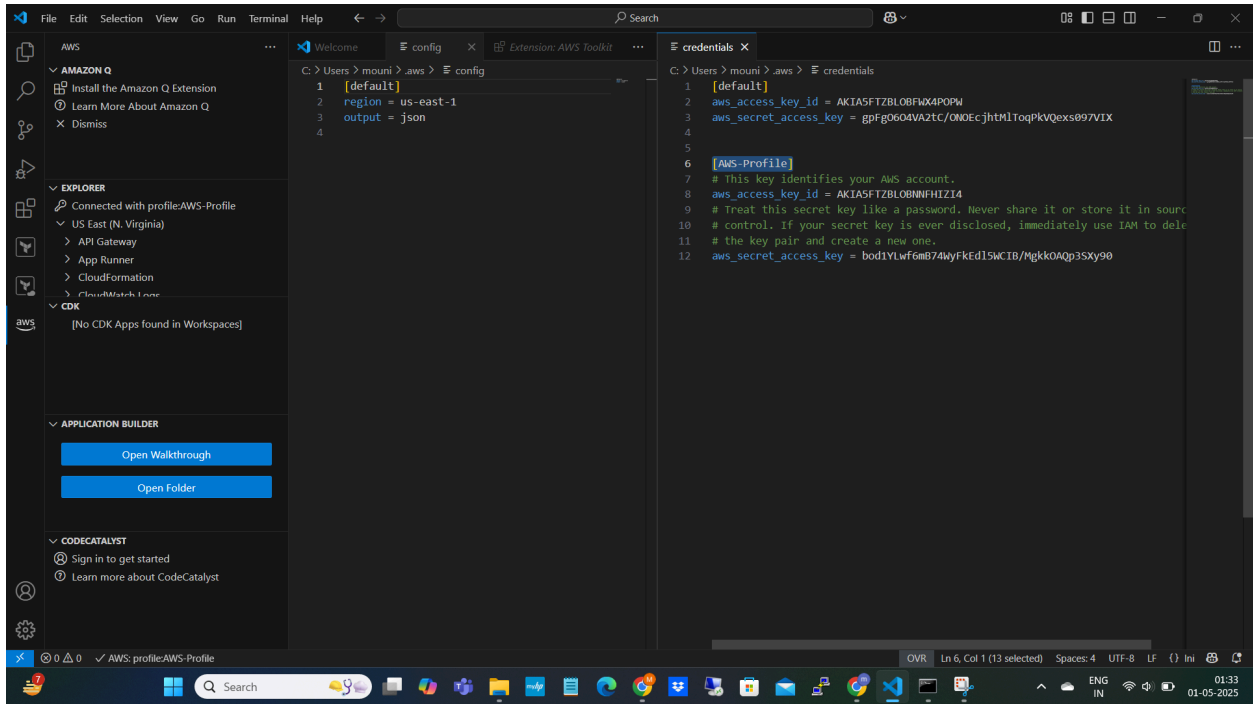


Enter Access key and Click enter



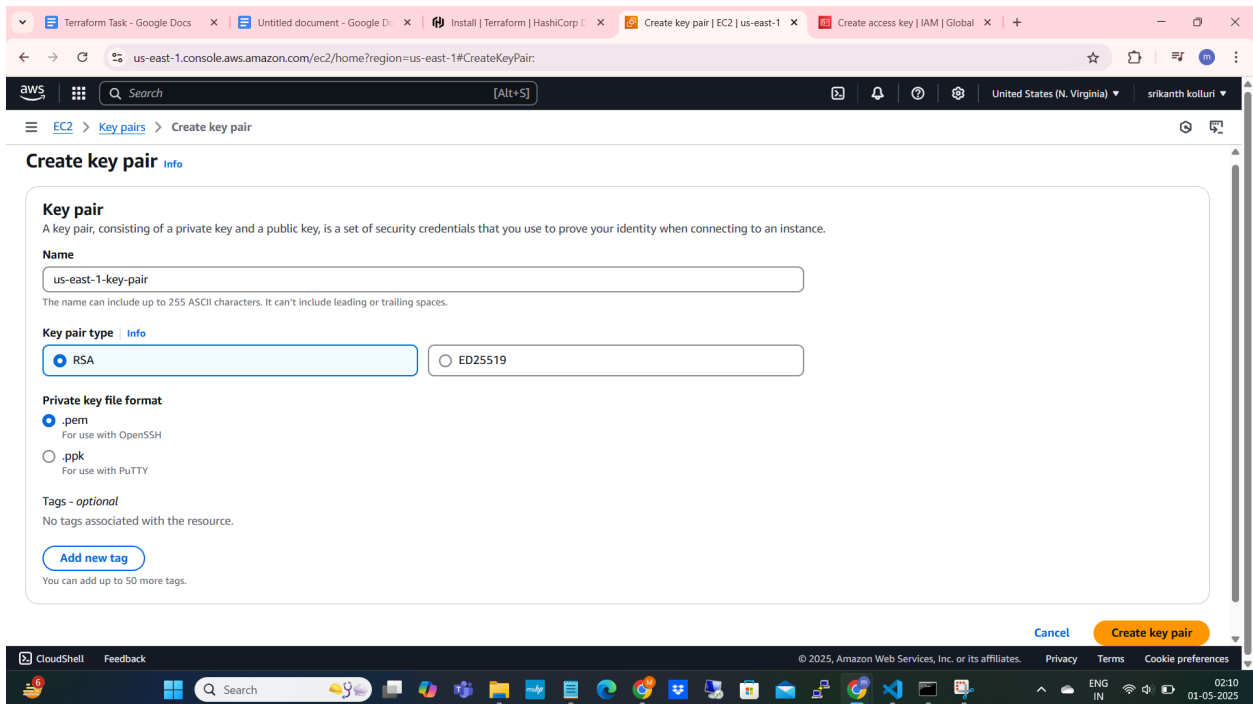
Enter the Secret Access key and click enter





It is Connected Now

Creating Key pairs in 2 different regions:
us-east-1-key-pair :



us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#KeyPairs:

Successfully created key pair

Key pairs (2) [Info](#) [Actions](#) [Create key pair](#)

Find Key Pair by attribute or tag

☐	Name	Type	Created	Fingerprint	ID
☐	us-east-1-key-pair	rsa	2025/05/01 02:11 GMT+5:30	75:40:2e:10:f6:7d:b5:2b:bd:07:e4:48:f9:3f:0c:...	key-075f1..
☐	aws-key	rsa	2025/04/08 22:58 GMT+5:30	5f:c1:36:c7:27:9f:2d:75:d0:be:8a:3e:6d:28:28:...	key-05532..

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02:11 01-05-2025

us-west-2-key-pair

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#KeyPairs:

Key pairs [Info](#) [Actions](#) [Create key pair](#)

Find Key Pair by attribute or tag

☐	Name	Type	Created	Fingerprint
--------------------------	------	------	---------	-------------

No key pairs to display

United States

- N. Virginia us-east-1
- Ohio us-east-2
- N. California us-west-1
- Oregon us-west-2**

Asia Pacific

- Mumbai ap-south-1
- Osaka ap-northeast-3
- Seoul ap-northeast-2
- Singapore ap-southeast-1
- Sydney ap-southeast-2
- Tokyo ap-northeast-1

Canada

- Central ca-central-1

Europe

- Frankfurt eu-central-1
- Ireland eu-west-1
- London eu-west-2
- Paris eu-west-3
- Stockholm eu-north-1

[Manage Regions](#) [Manage Local Zones](#)

https://us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#KeyPairs:

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02:11 01-05-2025

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#CreateKeyPair:

Create key pair [Info](#)

Key pair
A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.

Name

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type [Info](#)
☒ RSA ☐ ED25519

Private key file format
☒ .pem
For use with OpenSSH
☐ .ppk
For use with PuTTY

Tags - optional
No tags associated with the resource.
[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Create key pair](#)

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#KeyPairs:

Key pairs [Info](#)

Successfully created key pair

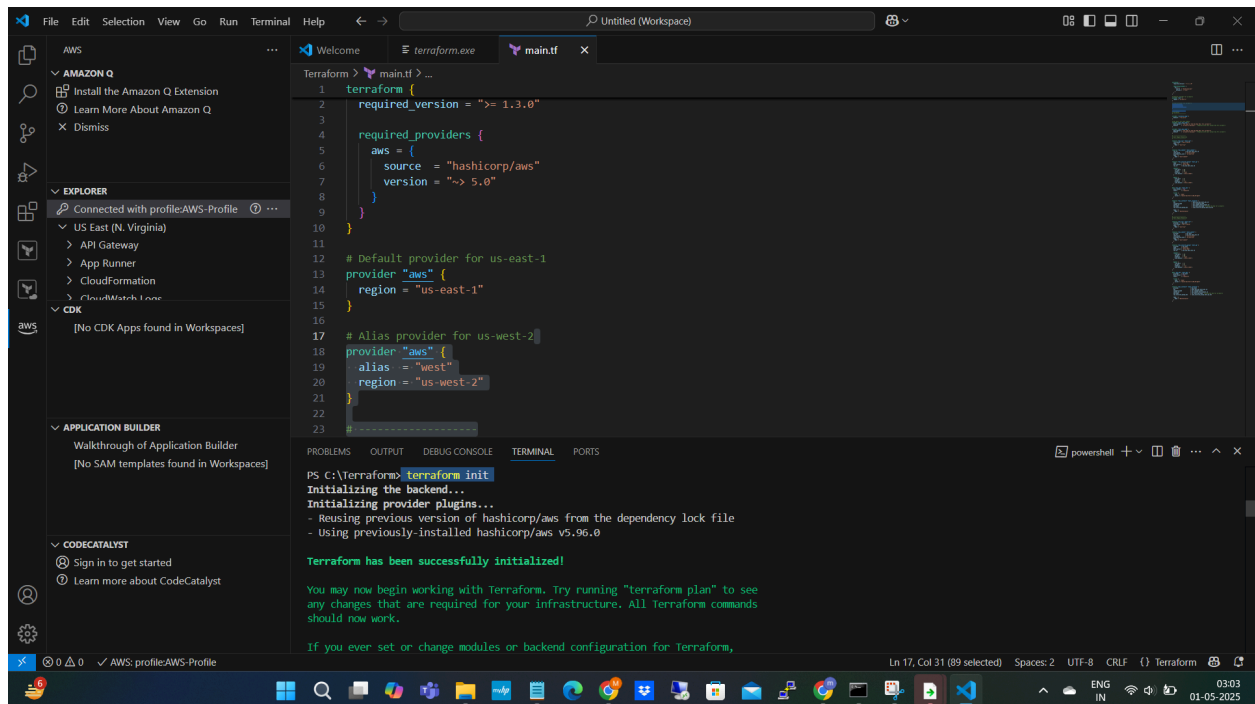
[Actions](#) [Create key pair](#)

☐	Name	Type	Created	Fingerprint	ID
☐	us-west-2-key-pair	rsa	2025/05/01 02:12 GMT+5:30	3f48:05:89:25:06:0e:88:3a:58:88:8c:f6:03:c9...	key-02caa...

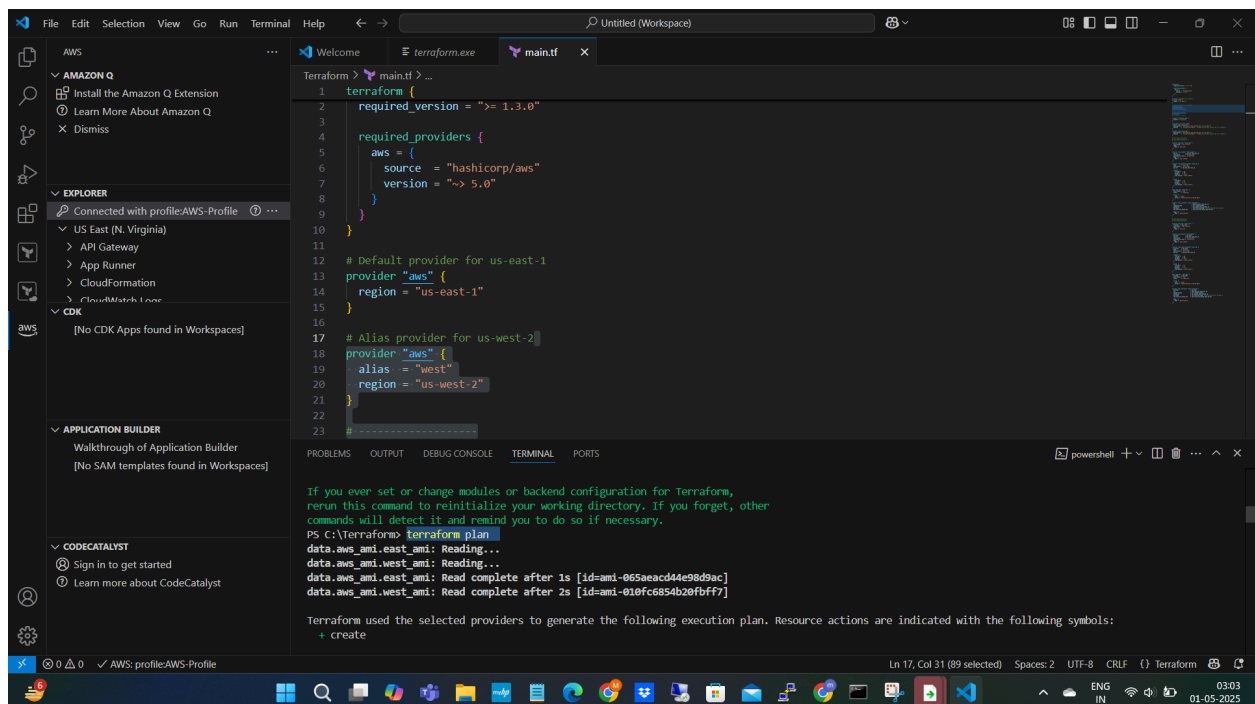
Added main.tf file to create 2 EC2 Instances in 2 Different Regions

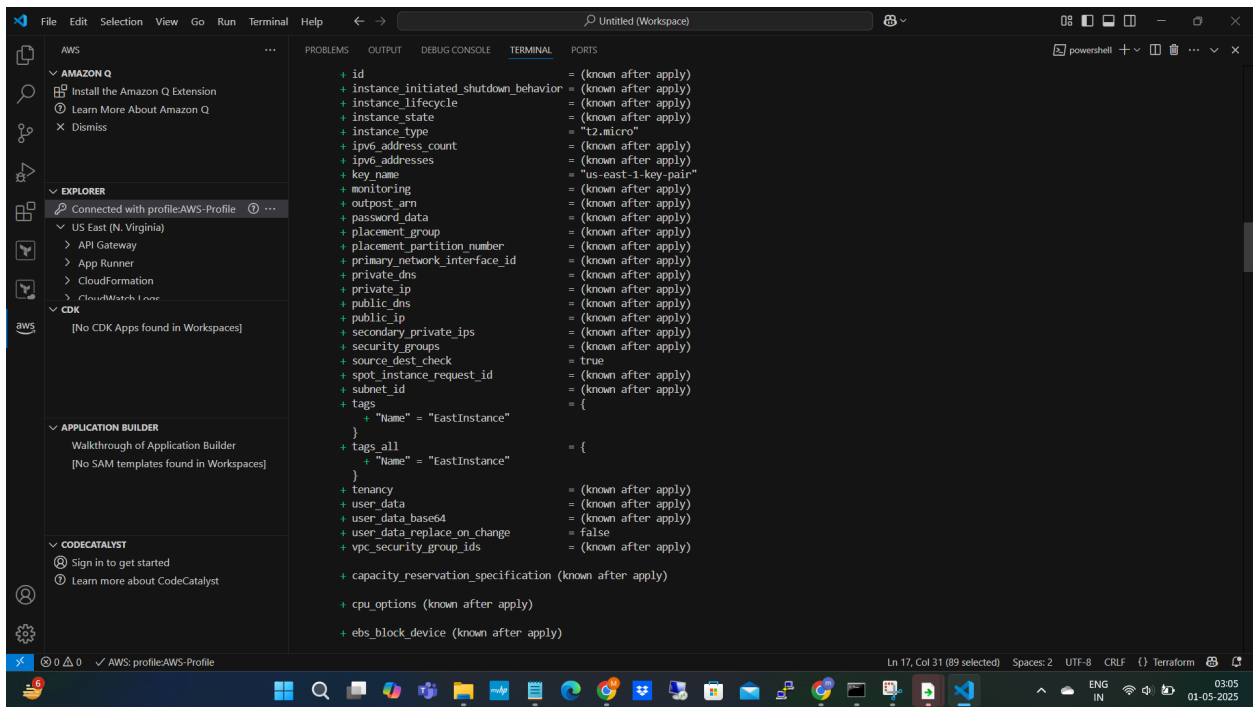
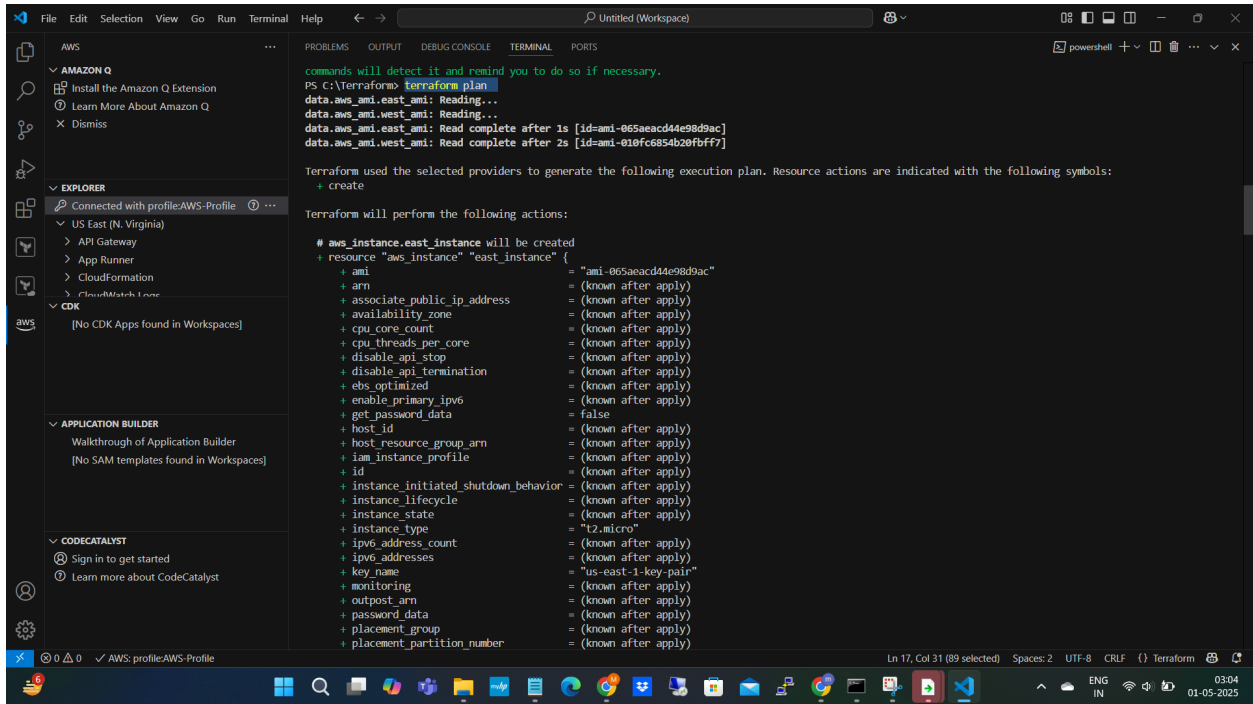
Now initializing the Terraform:

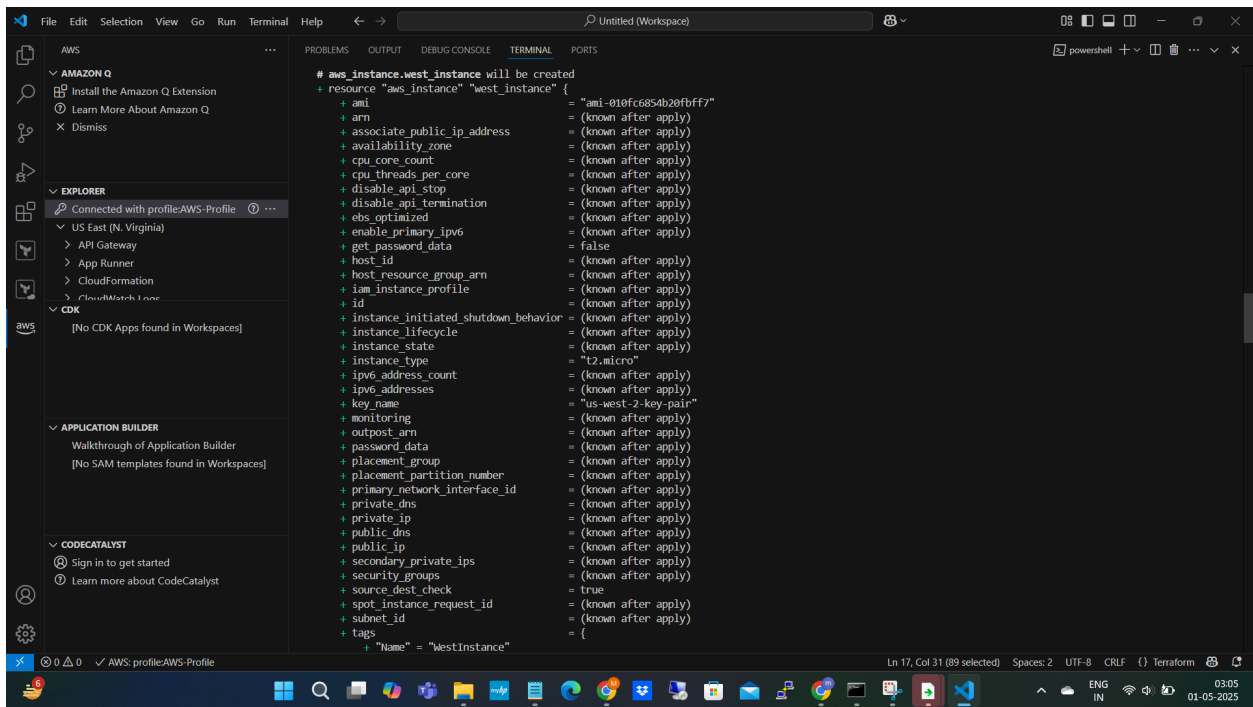
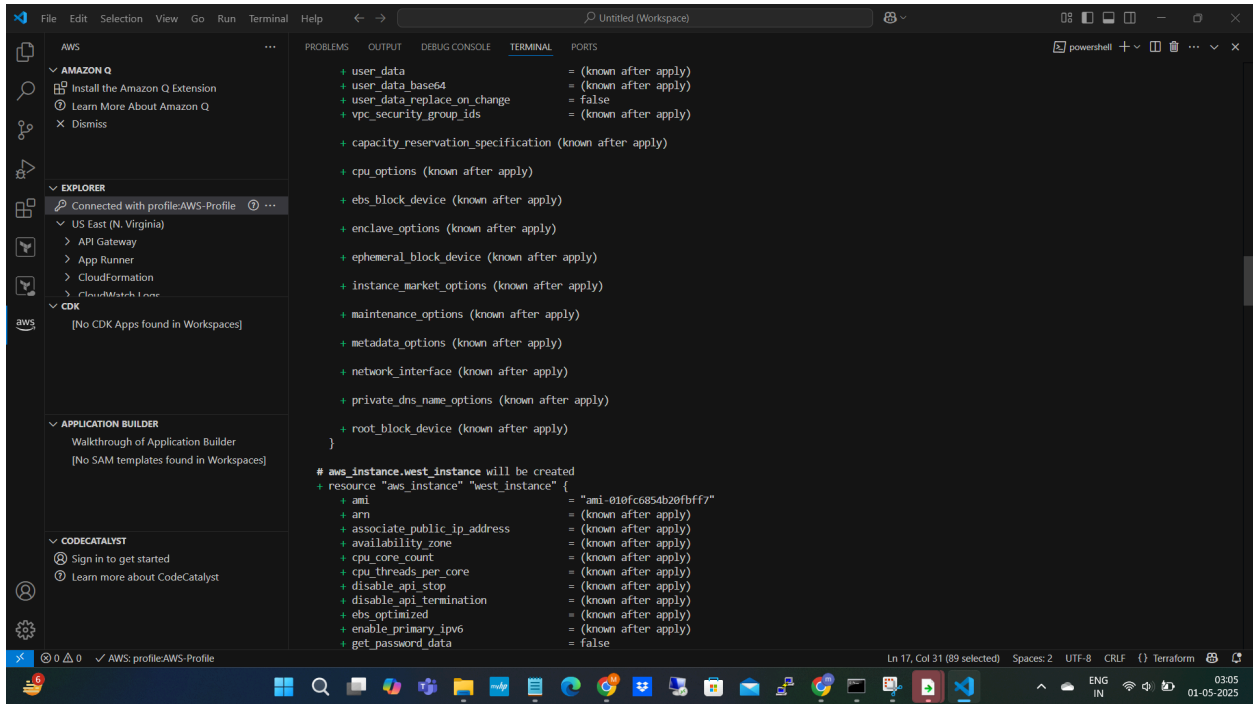
\$ terraform init

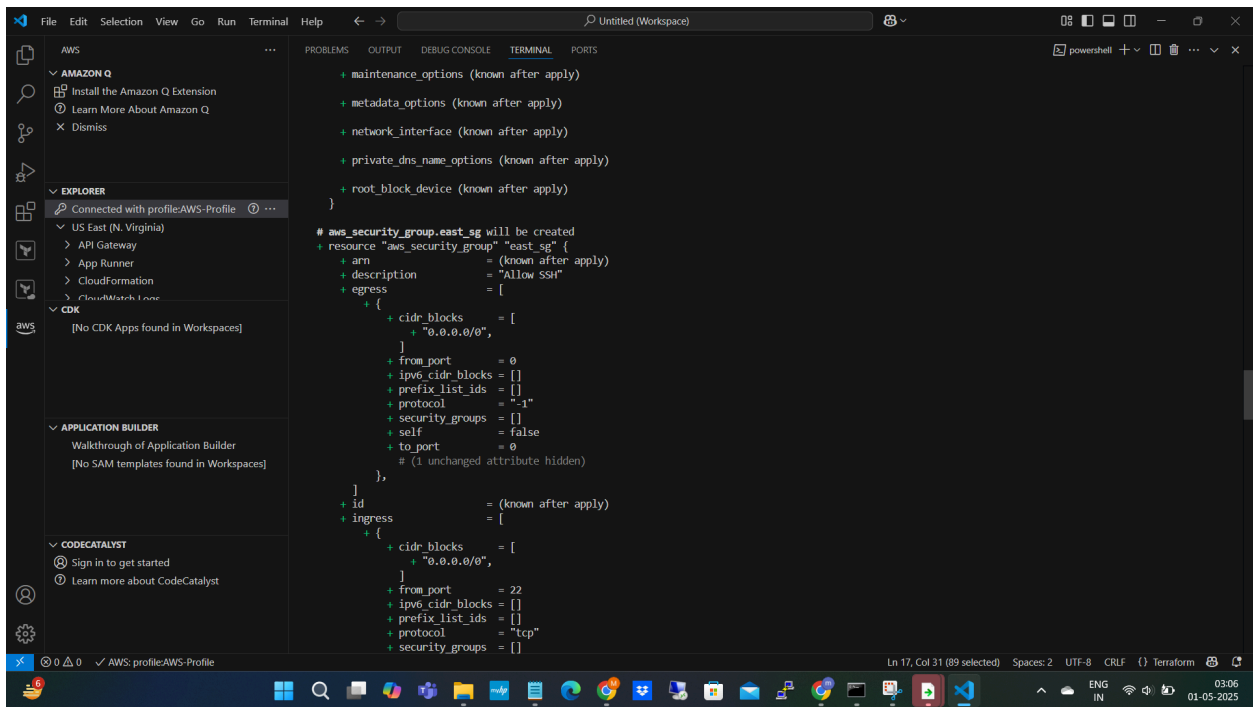
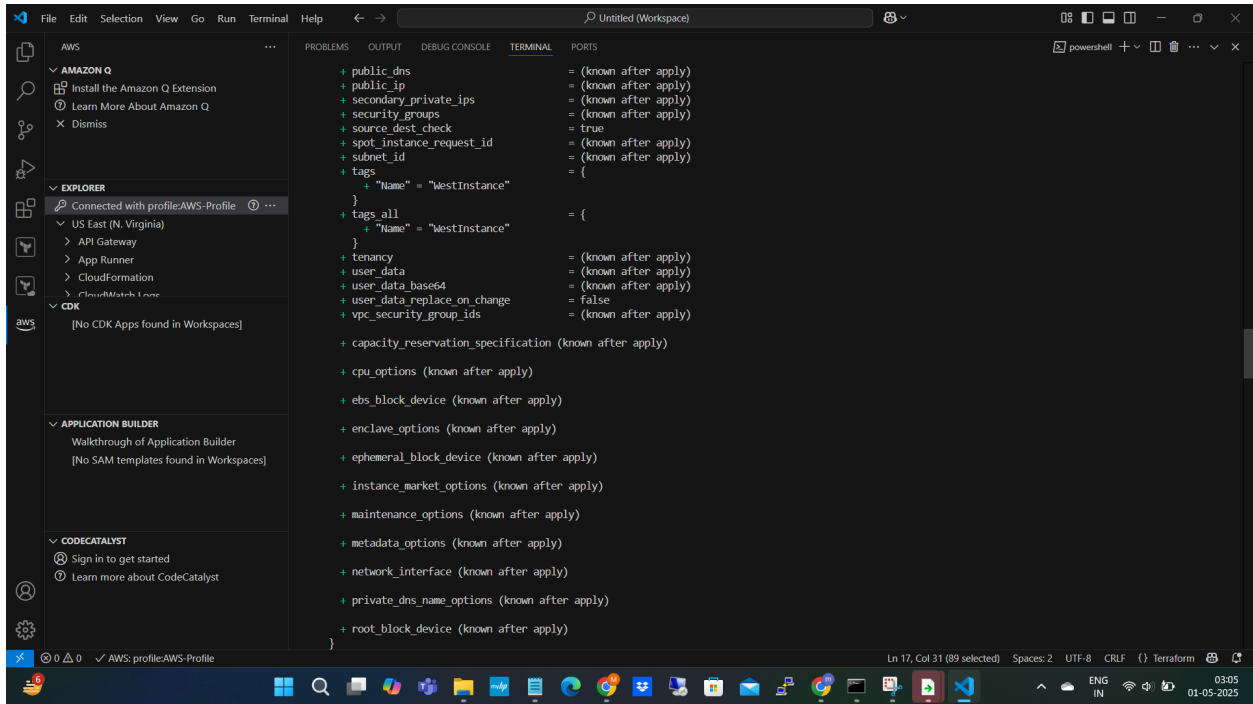


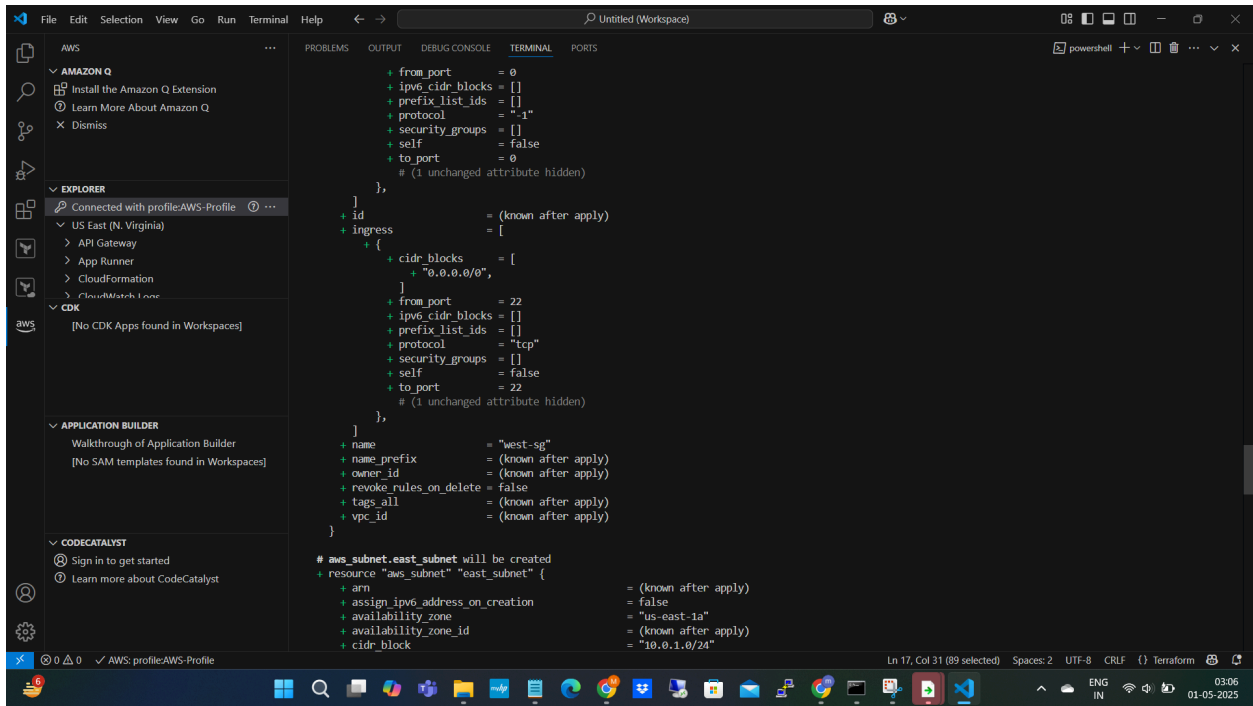
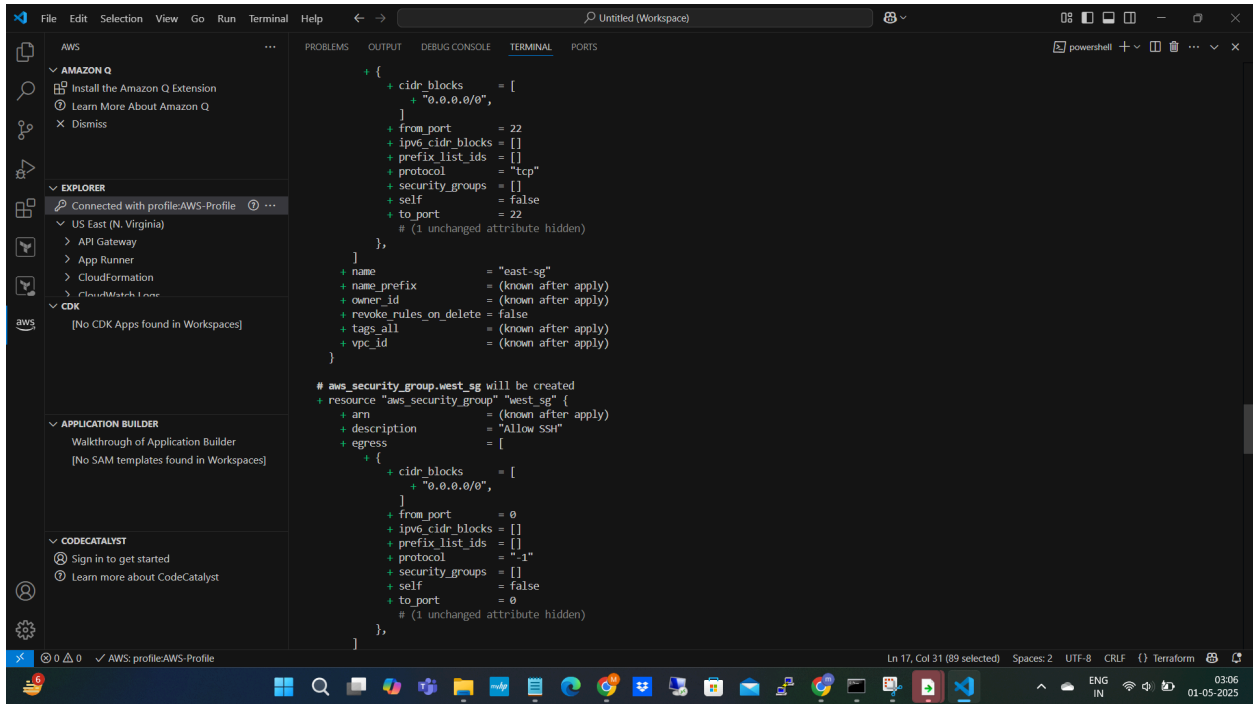
\$ terraform plan

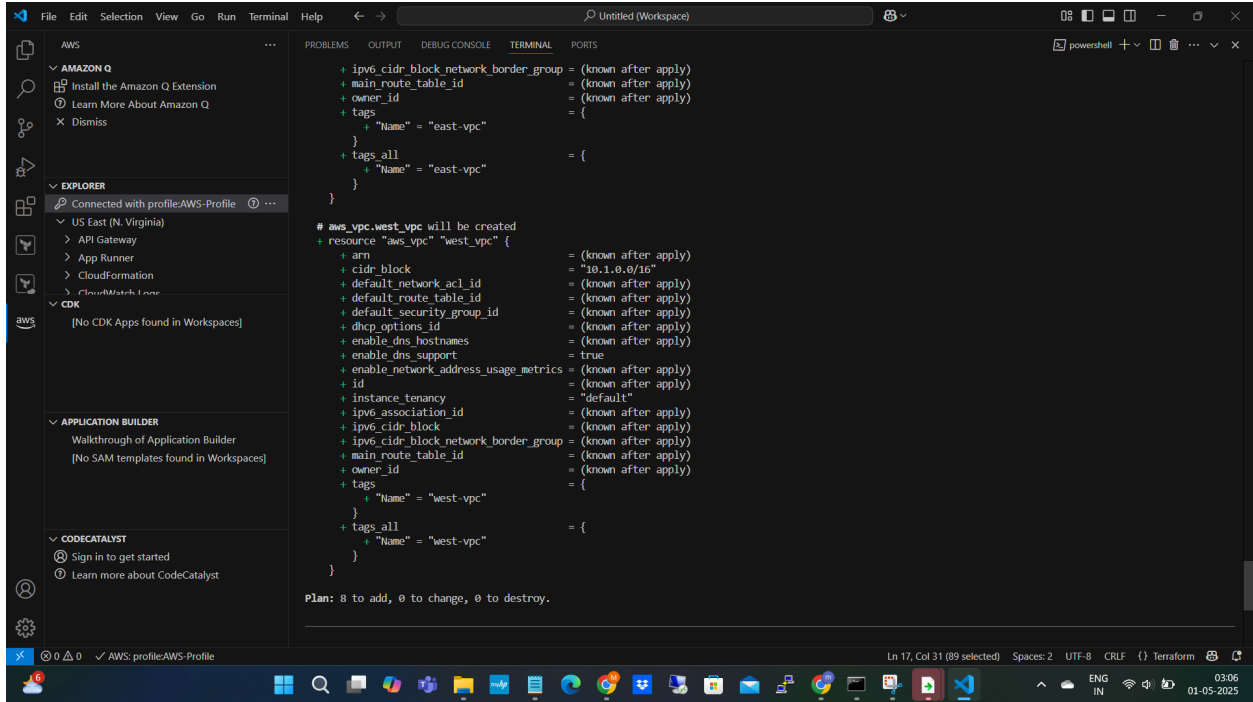




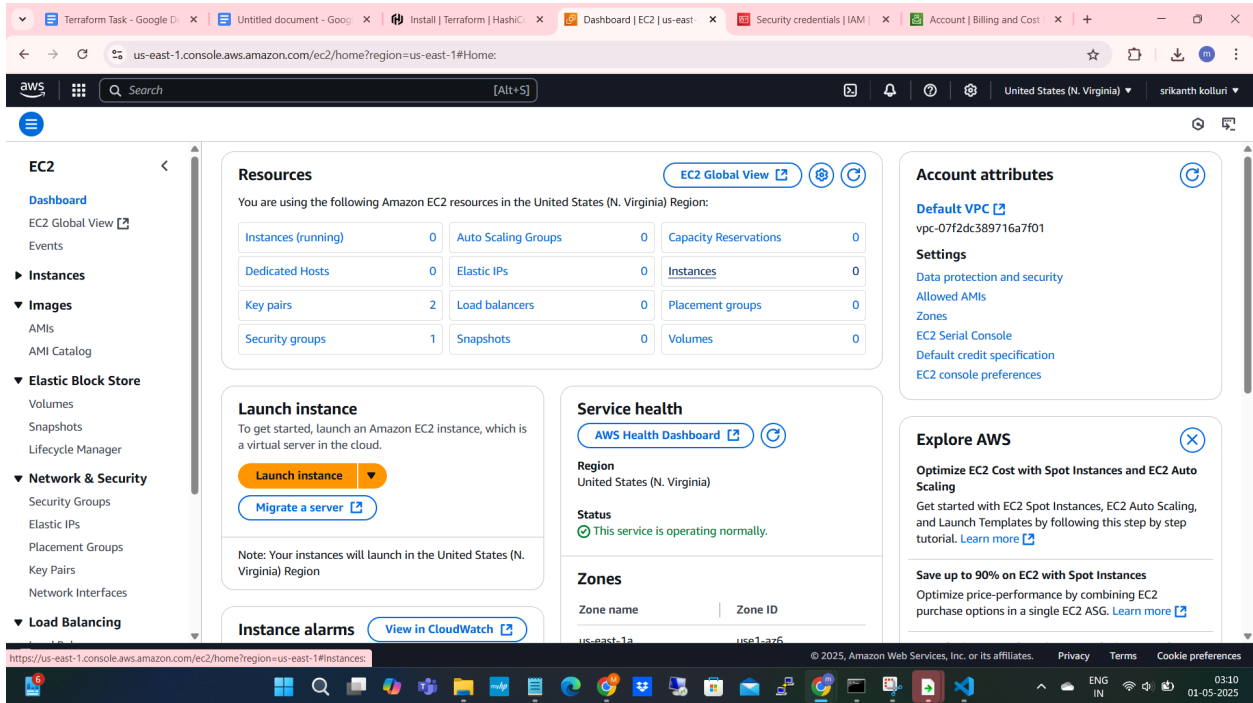




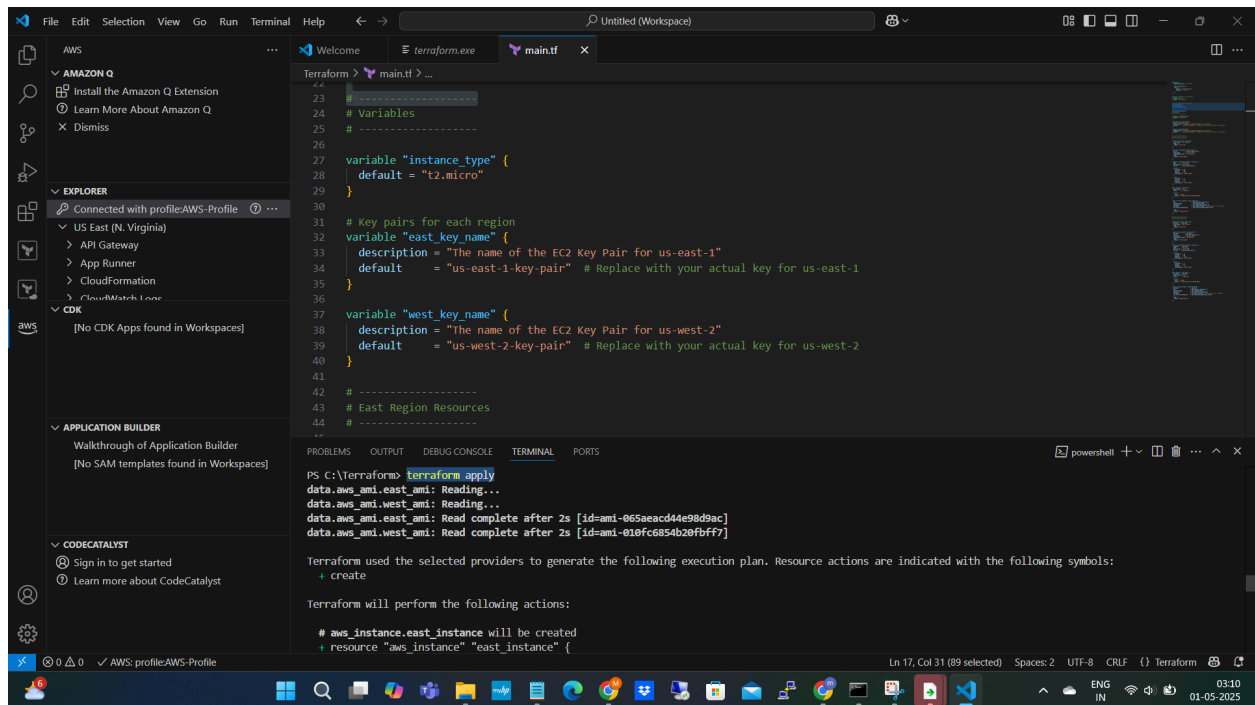




Before Apply- No Ec2 Instances



\$ terraform apply



The screenshot shows the Visual Studio Code interface with a Terraform configuration file named `main.tf` open. The file contains variables for instance type and key names, and resource definitions for EC2 instances in two regions. The terminal window shows the output of the `terraform apply` command, indicating that the resources are being created.

```
23 # -----
24 # Variables
25 # -----
26
27 variable "instance_type" {
28   default = "t2.micro"
29 }
30
31 # Key pairs for each region
32 variable "east_key_name" {
33   description = "The name of the EC2 Key Pair for us-east-1"
34   default     = "us-east-1-key-pair" # Replace with your actual key for us-east-1
35 }
36
37 variable "west_key_name" {
38   description = "The name of the EC2 Key Pair for us-west-2"
39   default     = "us-west-2-key-pair" # Replace with your actual key for us-west-2
40 }
41
42 # -----
43 # East Region Resources
44 # -----
```

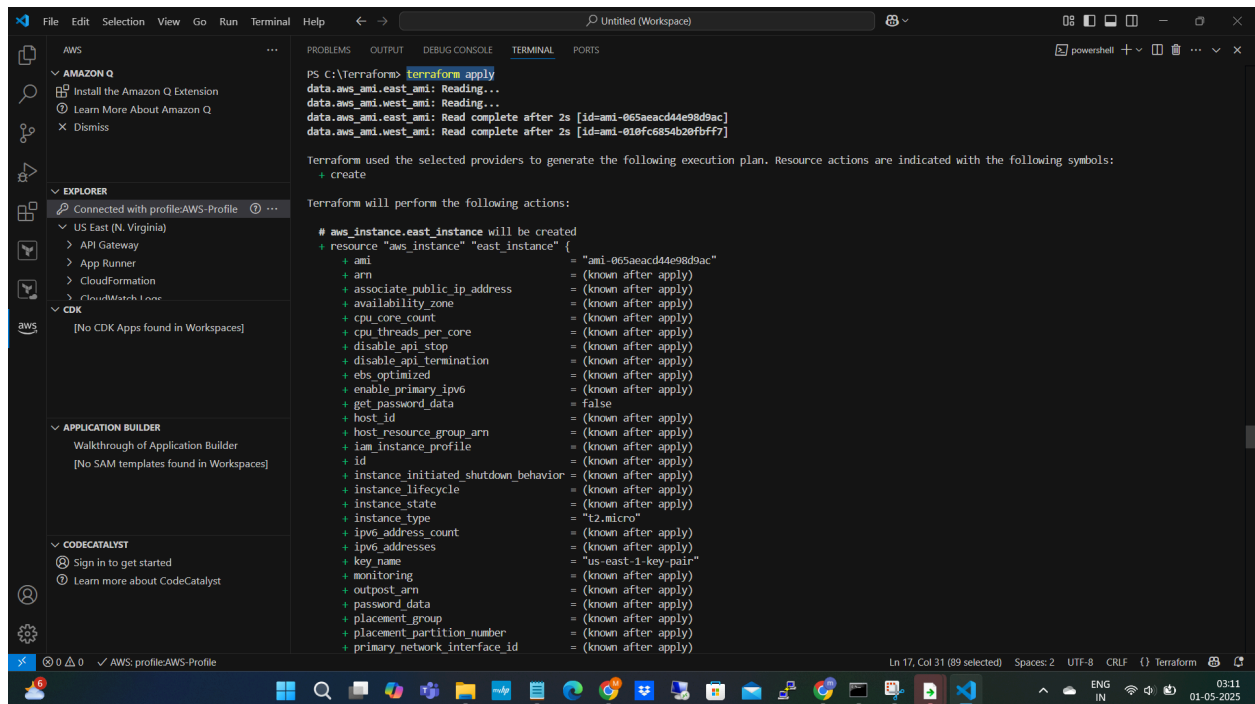
Terminal Output:

```
PS C:\Terraform> terraform apply
data.aws_ami.east_ami: Reading...
data.aws_ami.west_ami: Reading...
data.aws_ami.east_ami: Read complete after 2s [id=ami-065aeacd44e98d9ac]
data.aws_ami.west_ami: Read complete after 2s [id=ami-010fc6854b20fbff7]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

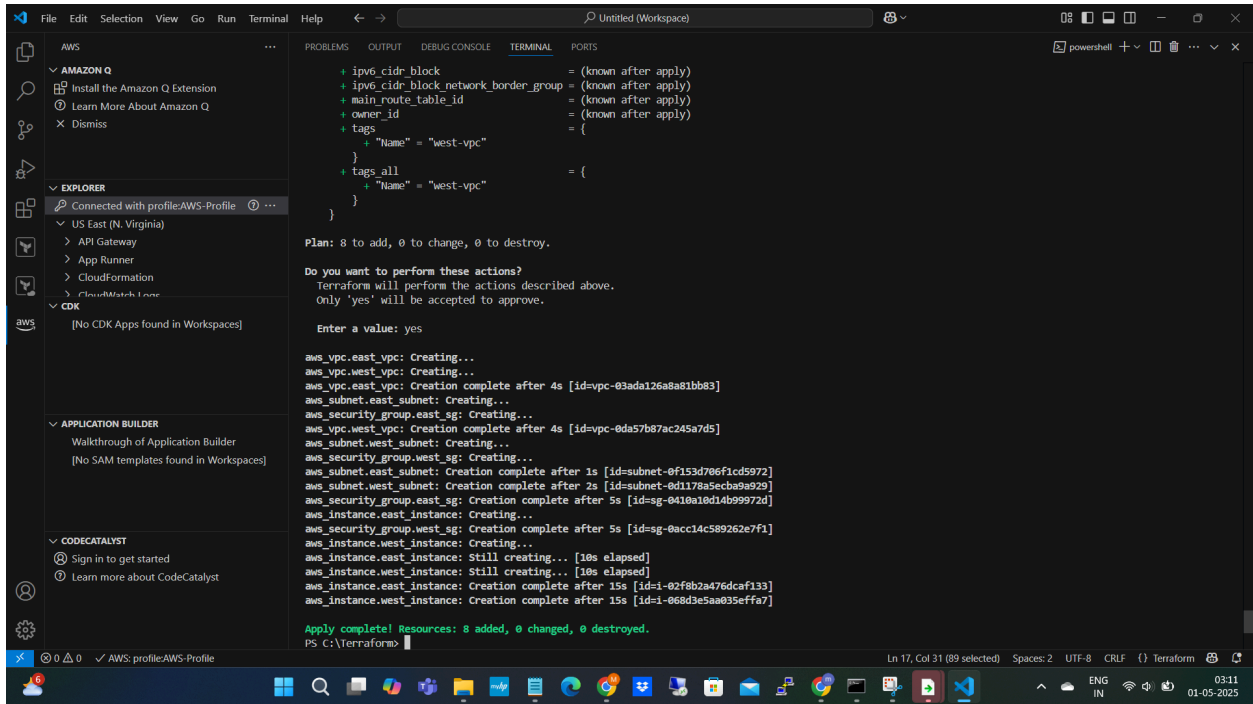
Terraform will perform the following actions:

# aws_instance.east_instance will be created
+ resource "aws_instance" "east_instance" {
```



This screenshot shows the continuation of the Terraform execution plan for the `aws_instance.east_instance` resource. The plan lists various attributes that will be created or updated during the apply process.

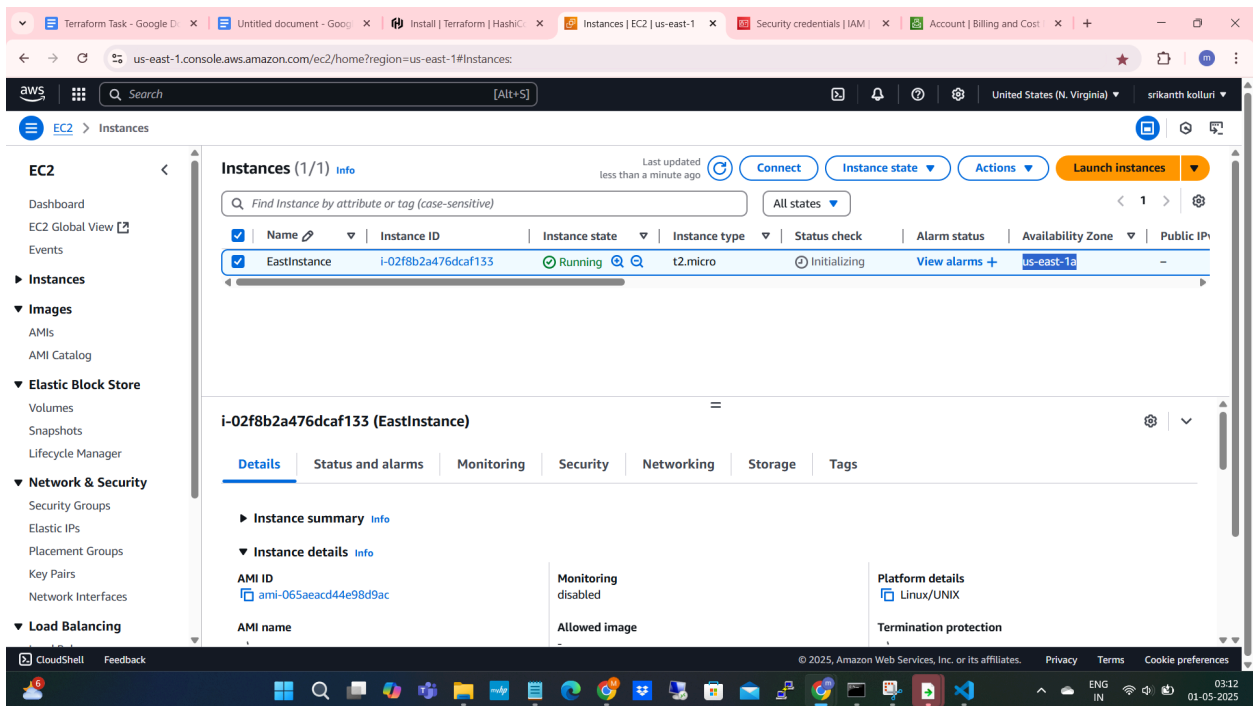
```
+ resource "aws_instance" "east_instance" {
+   ami               = "ami-065aeacd44e98d9ac"
+   arn               = (known after apply)
+   associate_public_ip_address = (known after apply)
+   availability_zone  = (known after apply)
+   cpu_core_count     = (known after apply)
+   cpu_threads_per_core = (known after apply)
+   disable_api_stop    = (known after apply)
+   disable_api_termination = (known after apply)
+   ebs_optimized       = (known after apply)
+   enable_primary_ipv6 = (known after apply)
+   get_password_data   = false
+   host_id             = (known after apply)
+   host_resource_group_arn = (known after apply)
+   iam_instance_profile = (known after apply)
+   id                 = (known after apply)
+   instance_initiated_shutdown_behavior = (known after apply)
+   instance_lifecycle = (known after apply)
+   instance_state      = (known after apply)
+   instance_type       = "t2.micro"
+   ipv6_address_count   = (known after apply)
+   ipv6_addresses       = (known after apply)
+   key_name            = "us-east-1-key-pair"
+   monitoring           = (known after apply)
+   outpost_arn         = (known after apply)
+   password_data        = (known after apply)
+   placement_group      = (known after apply)
+   placement_partition_number = (known after apply)
+   primary_network_interface_id = (known after apply)
```



Successfully applied the terraform

Check AWS whether the instances are created or not

Instance created in : us-east-1 region



us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetailsinstanceId=i-02f8b2a476dcf133

EC2 > Instances > i-02f8b2a476dcf133

Instance summary for i-02f8b2a476dcf133 (EastInstance) [Info](#)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-02f8b2a476dcf133	Public IPv4 address -	Private IPv4 addresses 10.0.1.231
IPV6 address -	Instance state Running	Public IPv4 DNS -
Hostname type IP name: ip-10-0-1-231.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-1-231.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation s. Learn more
Auto-assigned IP address -	VPC ID vpc-03ada126a8a81bb83 (east-vpc)	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-0f153d706f1cd5972 (east-subnet)	Managed false
IMDSv2 Optional ⚠ EC2 recommends setting IMDSv2 to required Learn more	Instance ARN arn:aws:ec2:us-east-1:905418201986:instance/i-02f8b2a476dcf133	

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03:13 01-05-2025

Instance created in : us-west-2 region

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#Instances:

EC2 > Instances

Instances (1/1) [Info](#)

Last updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☒	WestInstance	i-068d3e5aa035effa7	Running	t2.micro	2/2 checks passed	View alarms	us-west-2a	-

i-068d3e5aa035effa7 (WestInstance)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary [Info](#)

Instance ID i-068d3e5aa035effa7	Public IPv4 address -	Private IPv4 addresses 10.1.1.61
IPV6 address -	Instance state Running	Public IPv4 DNS -
Hostname type	Private IP DNS name (IPv4 only)	

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03:14 01-05-2025

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#InstanceDetails:instanceId=i-068d3e5aa035effa7

EC2 > Instances > i-068d3e5aa035effa7

Instance summary for i-068d3e5aa035effa7 (WestInstance) [Info](#)

Updated less than a minute ago

Instance ID i-068d3e5aa035effa7	Public IPv4 address -	Private IPv4 addresses 10.1.1.61
IPv6 address -	Instance state Running	Public IPv4 DNS -
Hostname type IP name: ip-10-1-1-61.us-west-2.compute.internal	Private IP DNS name (IPv4 only) ip-10-1-1-61.us-west-2.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation Learn more
Auto-assigned IP address -	VPC ID vpc-0da57b87ac245a7d5 (west-vpc)	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-0d1178a5ecba9a929 (west-subnet)	Managed false
IMDSv2 Optional ⚠ EC2 recommends setting IMDSv2 to required Learn more	Instance ARN arn:aws:ec2:us-west-2:905418201986:instance/i-068d3e5aa035effa7	

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ENG IN 03:15 01-05-2025

Both EC2 instances created in us-east-1 and us-east-2 regions with the single terraform